

Longitudinal Clinical Intelligence & Care Coordination Platform

Principal AI Architect Architecture Case Study

ROLE	Principal AI Architect (AI Platforms & Distributed Systems)
DOMAIN	Integrated Healthcare Network — USA
CLOUD	Microsoft Azure
DELIVERY	20-month multi-phase platform • 22-member cross-functional engineering team

Architecture position

Establish a governed path from fragmented clinical data to longitudinal patient context, evidence-grounded retrieval, assistive AI reasoning and clinician-controlled workflow action — without allowing model output to become the authoritative clinical decision boundary.

Production evidence: 10+ workflow domains • sub-second context/retrieval • 30%+ less manual context consolidation • 25%+ lower context-processing latency • 35%+ less redundant LLM inference

Executive Architecture Mandate

The architecture is organized around responsibility boundaries rather than a technology inventory. The target architecture establishes explicit seams for interoperability, event propagation, canonical clinical context, evidence retrieval, AI execution, trust controls, workflow authority and production operations.

Architectural invariant: clinical source data → governed longitudinal context → evidence retrieval → bounded AI reasoning → clinician-controlled workflow action.

Documented implementation baseline: FHIR/HL7, Azure API Management, Azure Event Hubs, Patient Context Fabric, Azure AI Search, model/inference abstraction, adaptive context assembly, semantic caching, token optimization, Azure Content Safety, Presidio, AI Evaluation / LLM-as-a-Judge, AKS, Bicep, Azure DevOps, OpenTelemetry, Azure Monitor and Application Insights.

Architecture Evidence Map

The evidence map connects business intent, constraints, architecture boundaries and measurable production behavior. Outcomes are treated as validation evidence rather than decorative KPIs.

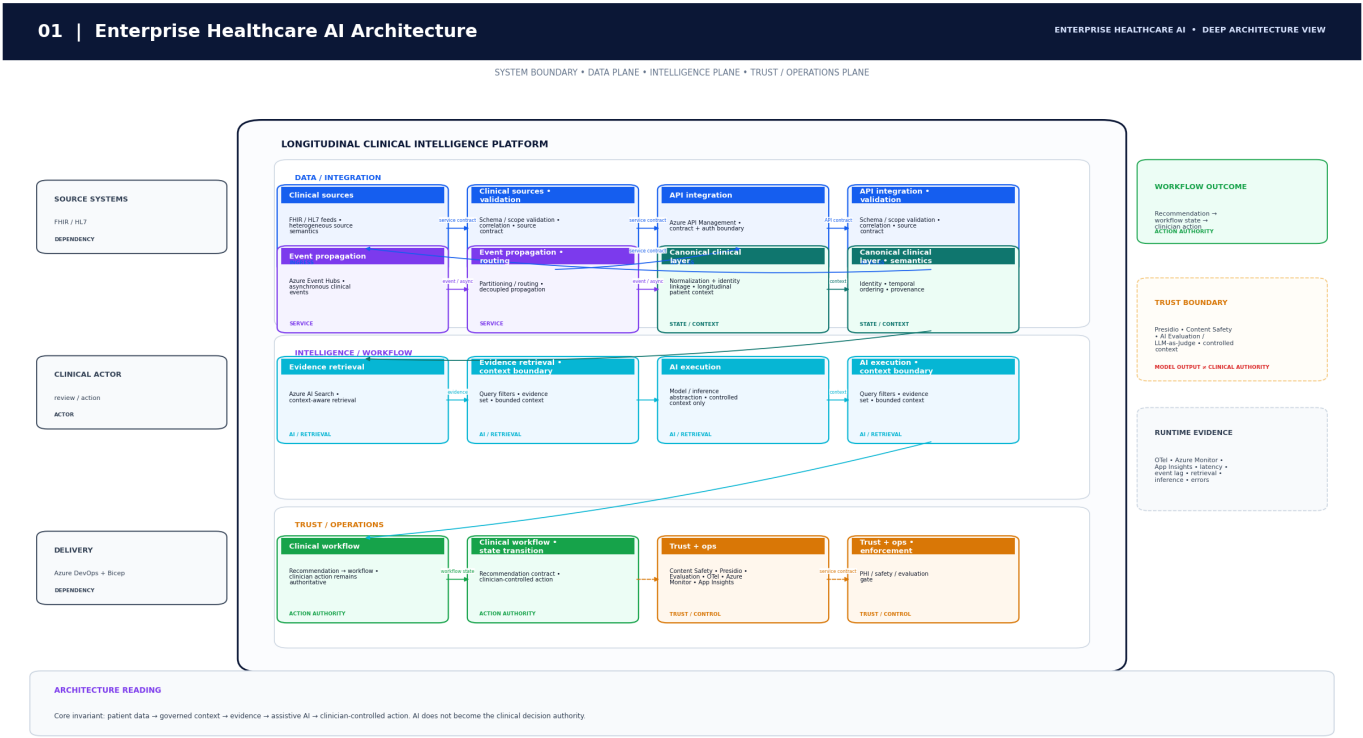
- 10+ clinical workflow domains
- Sub-second clinical-context / retrieval availability
- 30%+ reduction in manual context consolidation
- 25%+ reduction in context-processing latency
- 35%+ reduction in redundant LLM inference

Architecture decision standard

Every significant boundary is evaluated through responsibility ownership, latency, reliability, safety, operational measurability, reuse and cost. The portfolio separates documented production evidence from architecture control patterns.

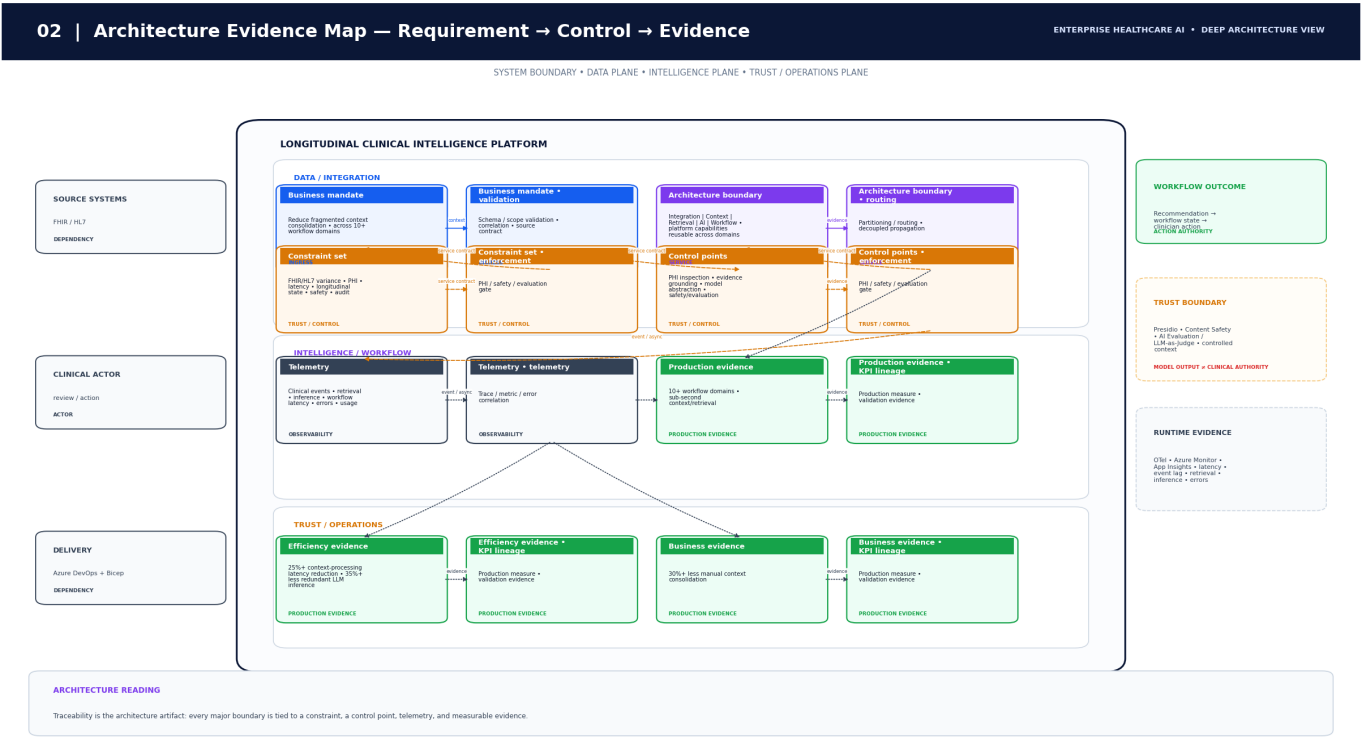
01 | Enterprise Healthcare AI Architecture

Deep architecture view



02 | Architecture Evidence Map — Requirement → Control → Evidence

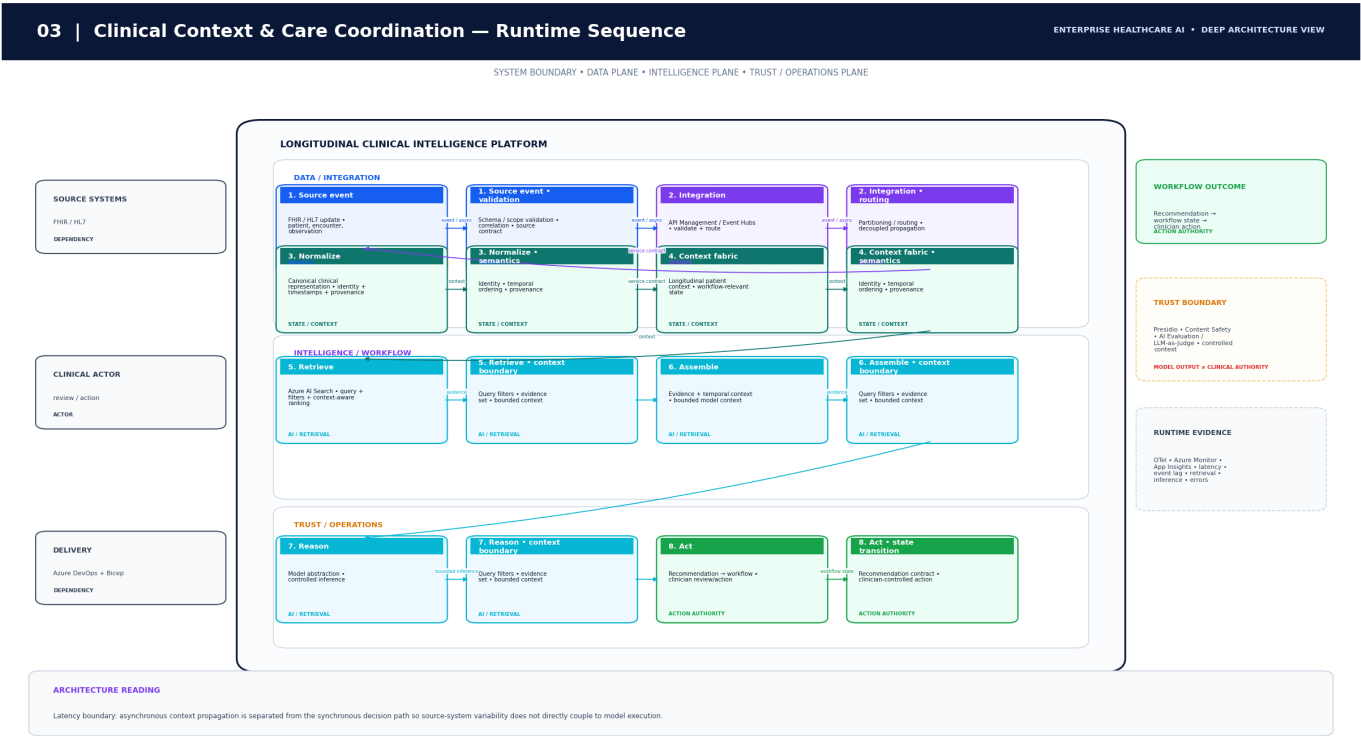
Deep architecture view



Architecture reading: Traceability is the architecture artifact: every major boundary is tied to a constraint, a control point, telemetry, and measurable evidence.

03 | Clinical Context & Care Coordination — Runtime Sequence

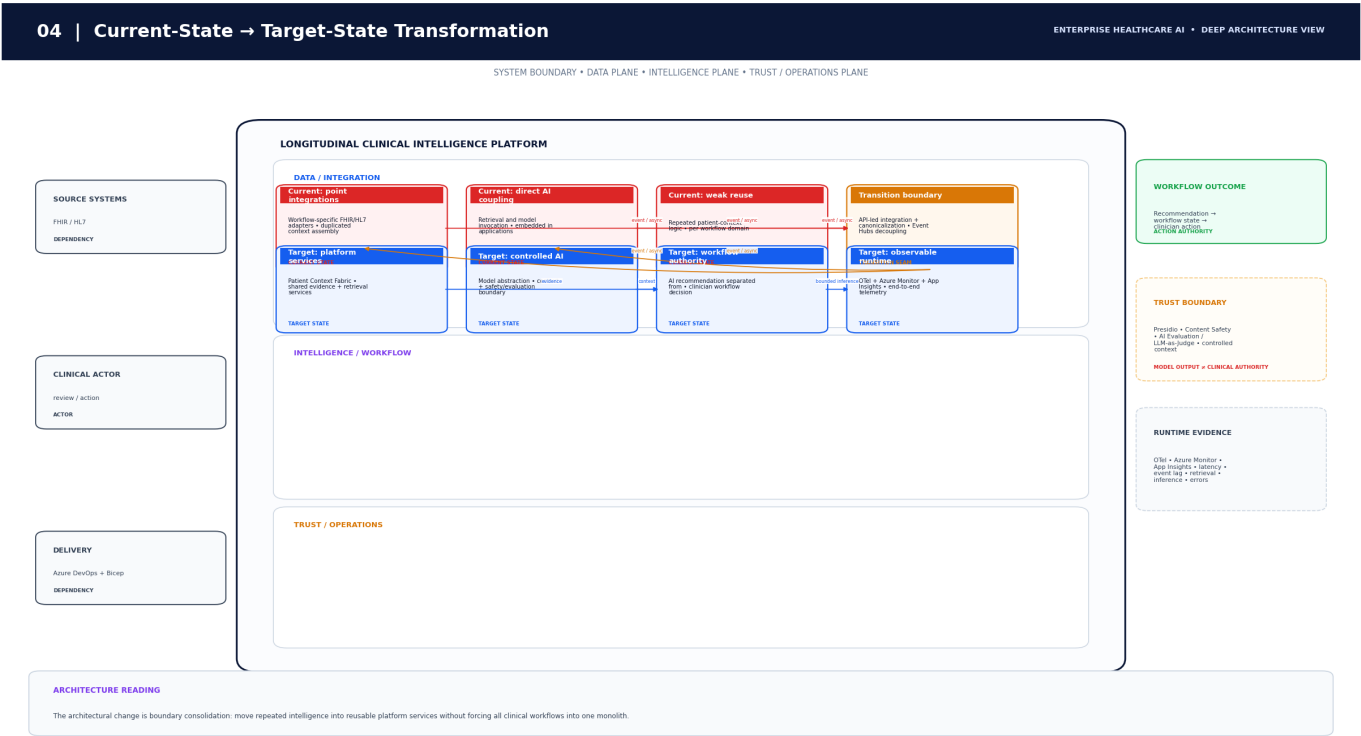
Deep architecture view



Architecture reading: Latency boundary: asynchronous context propagation is separated from the synchronous decision path so source-system variability does not directly couple to model execution.

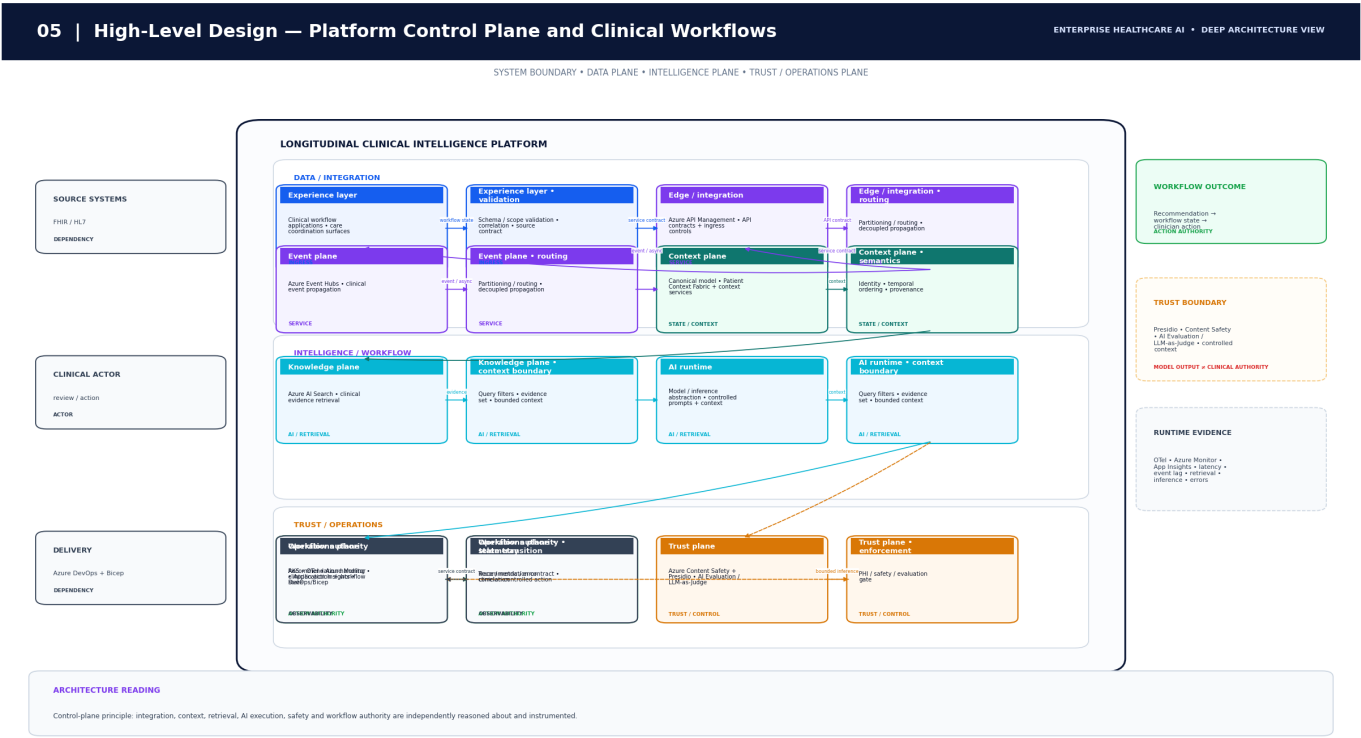
04 | Current-State → Target-State Transformation

Deep architecture view



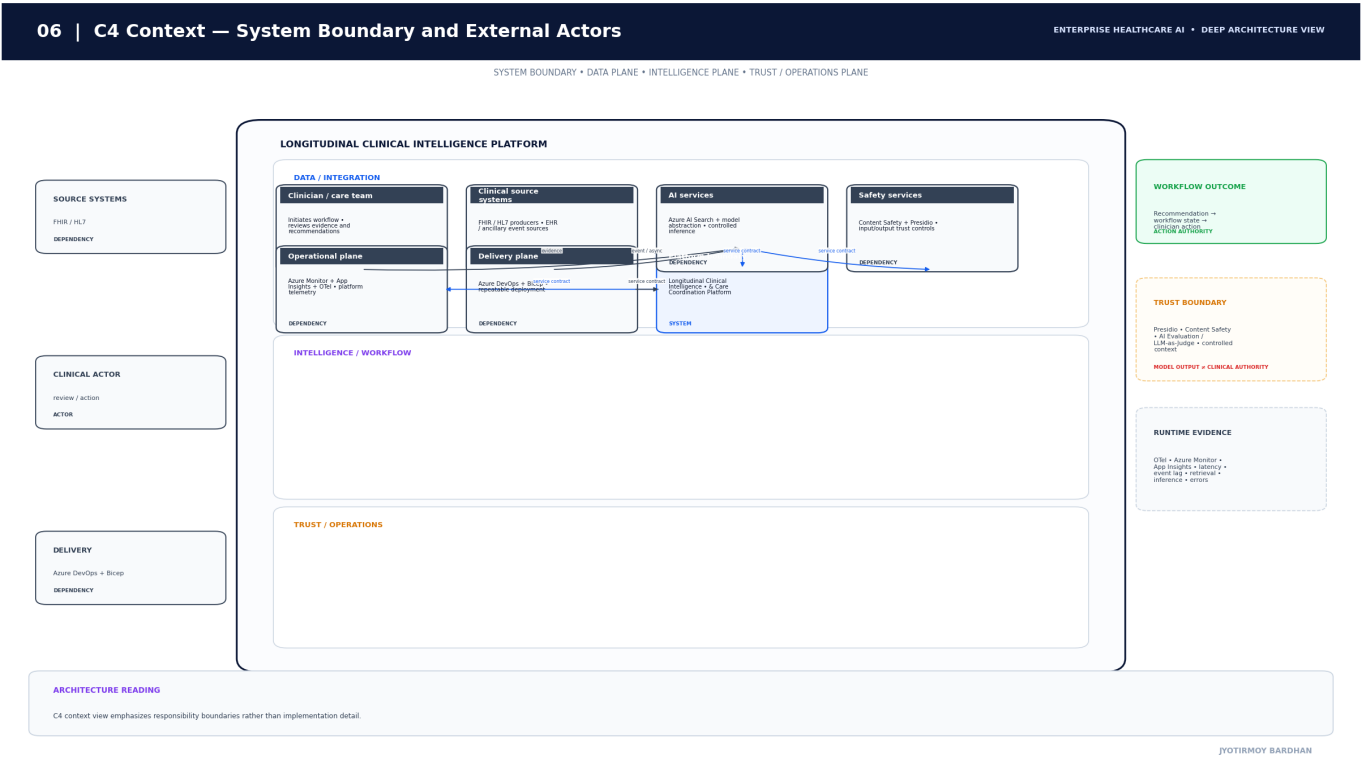
05 | High-Level Design — Platform Control Plane and Clinical Workflows

Deep architecture view



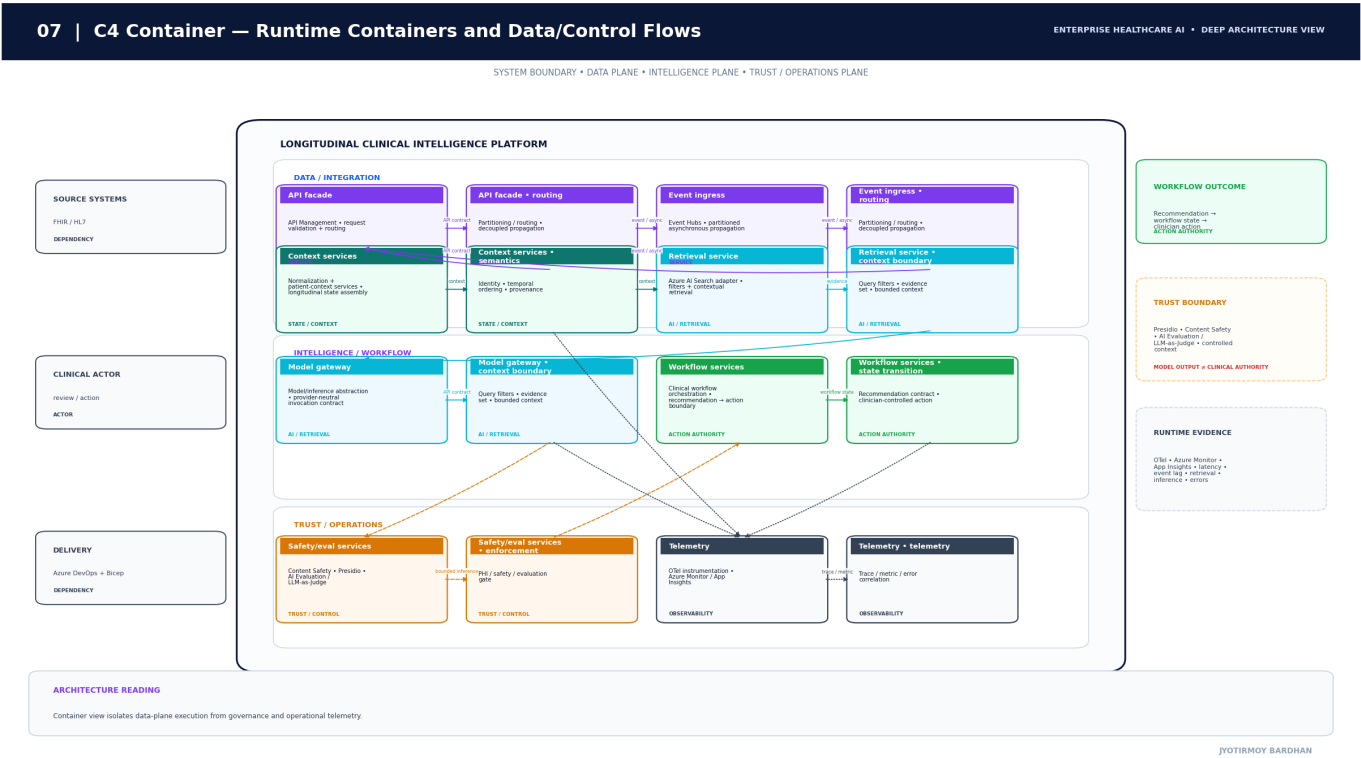
06 | C4 Context — System Boundary and External Actors

Deep architecture view



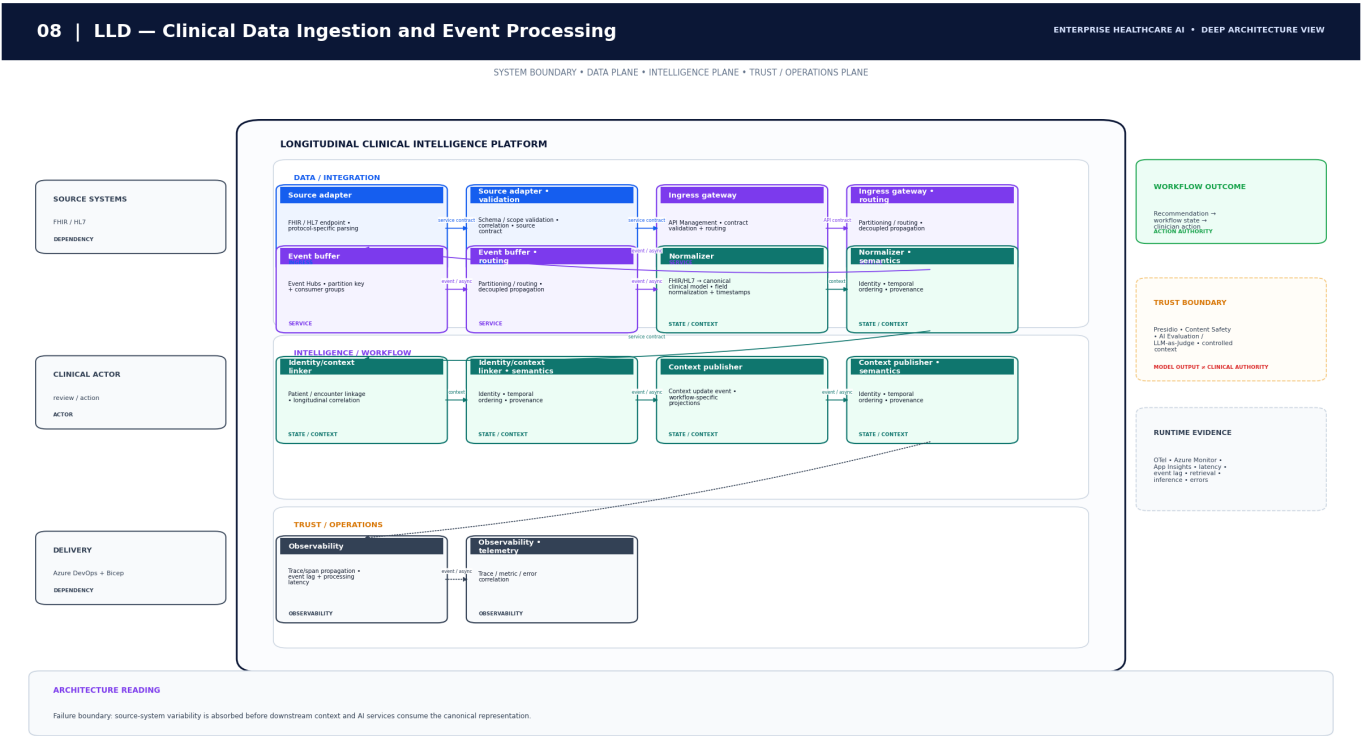
07 | C4 Container — Runtime Containers and Data/Control Flows

Deep architecture view



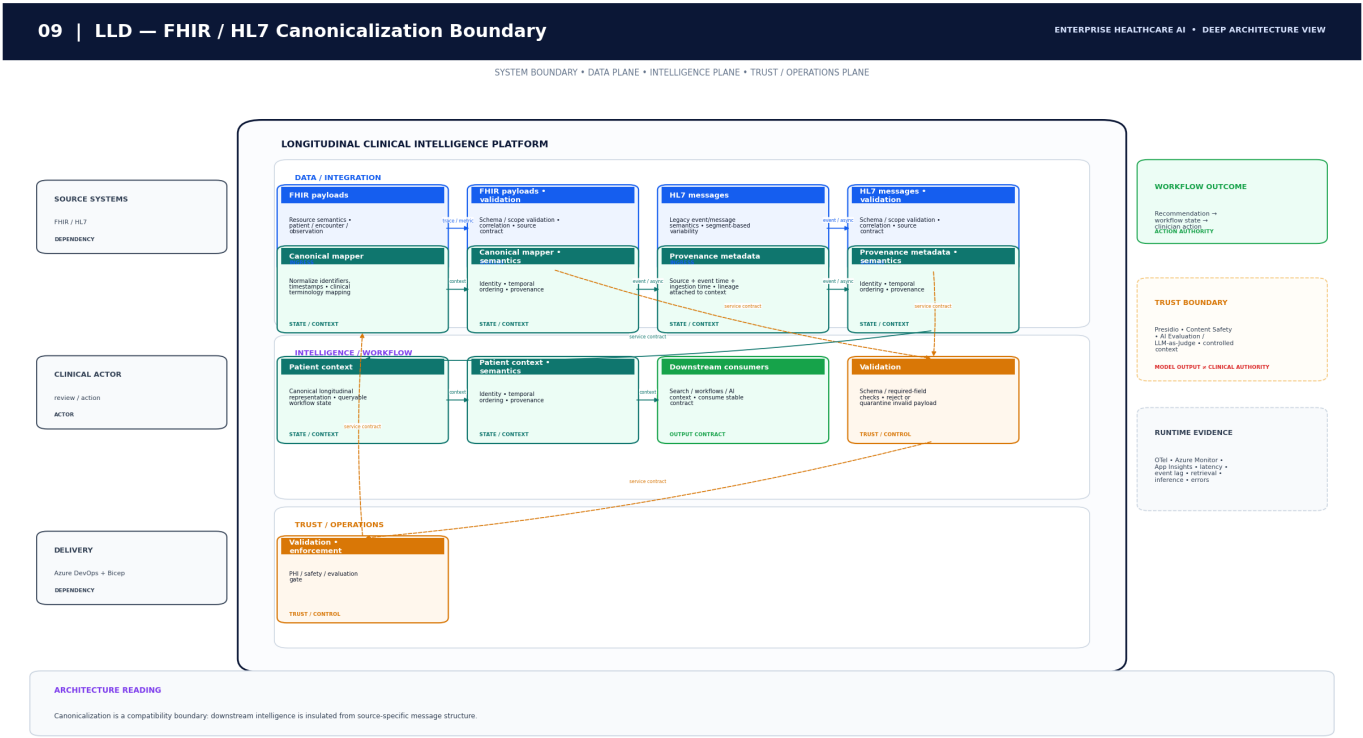
08 | LLD — Clinical Data Ingestion and Event Processing

Deep architecture view



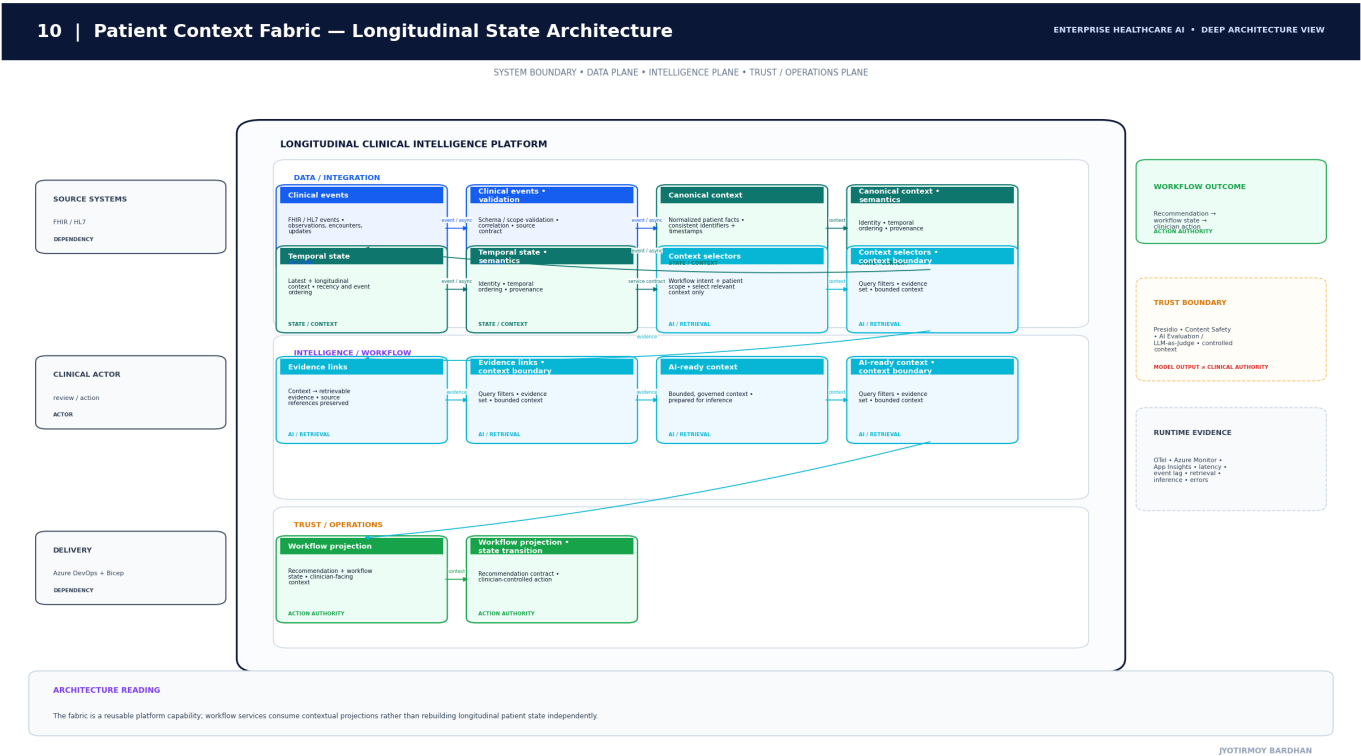
09 | LLD — FHIR / HL7 Canonicalization Boundary

Deep architecture view



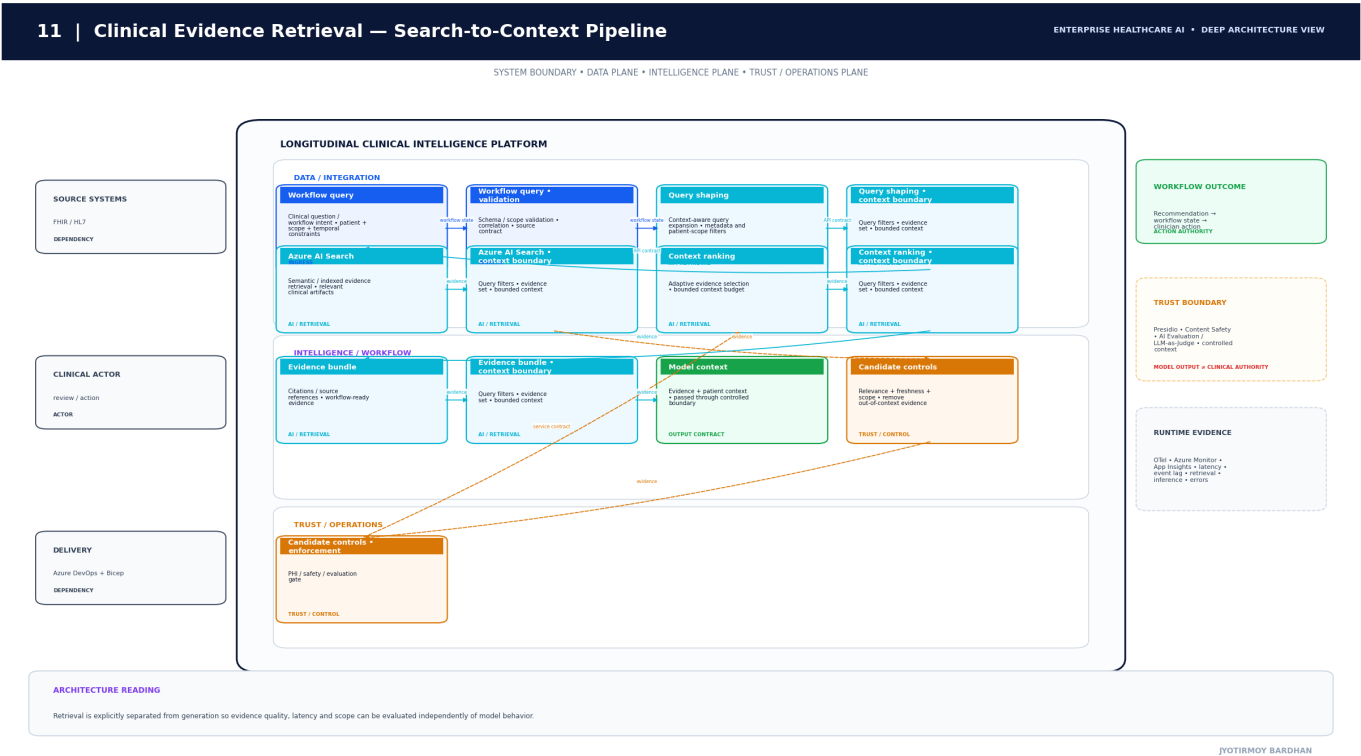
10 | Patient Context Fabric — Longitudinal State Architecture

Deep architecture view



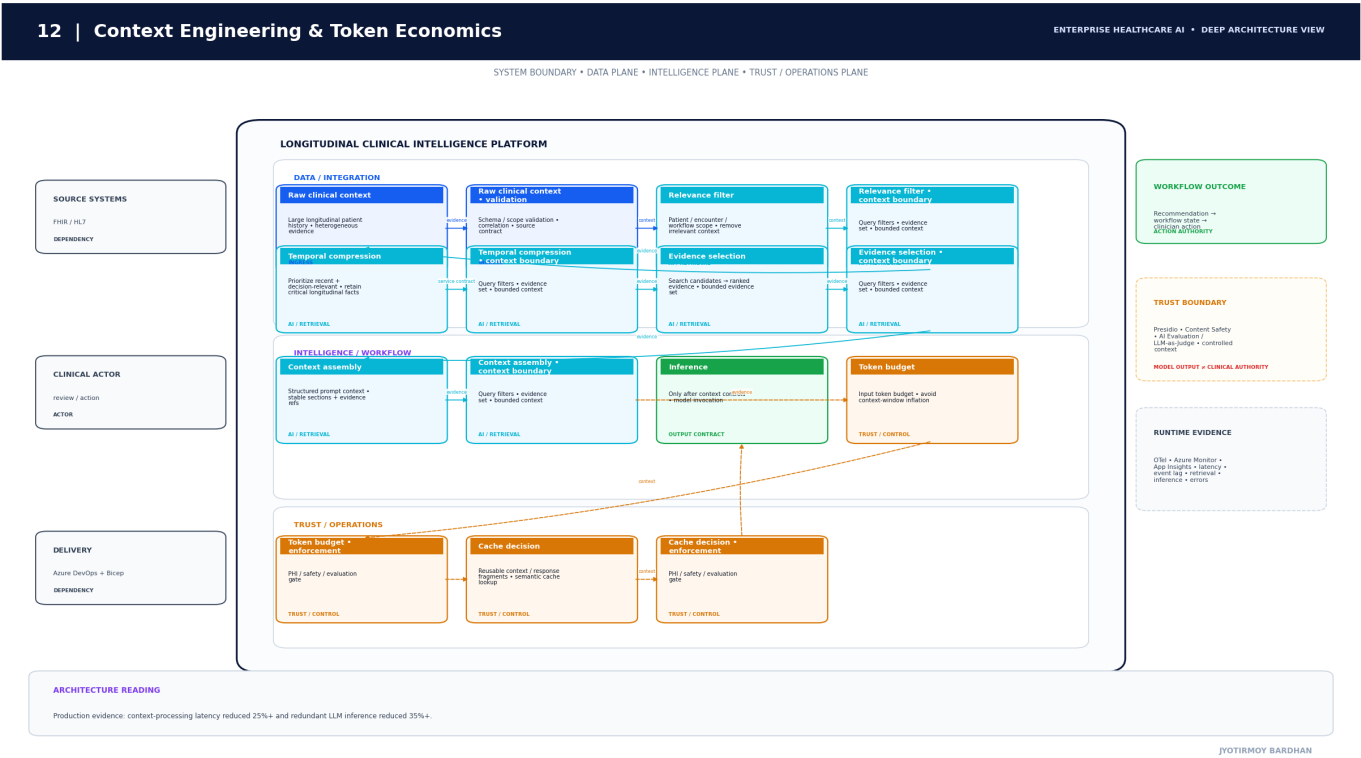
11 | Clinical Evidence Retrieval — Search-to-Context Pipeline

Deep architecture view



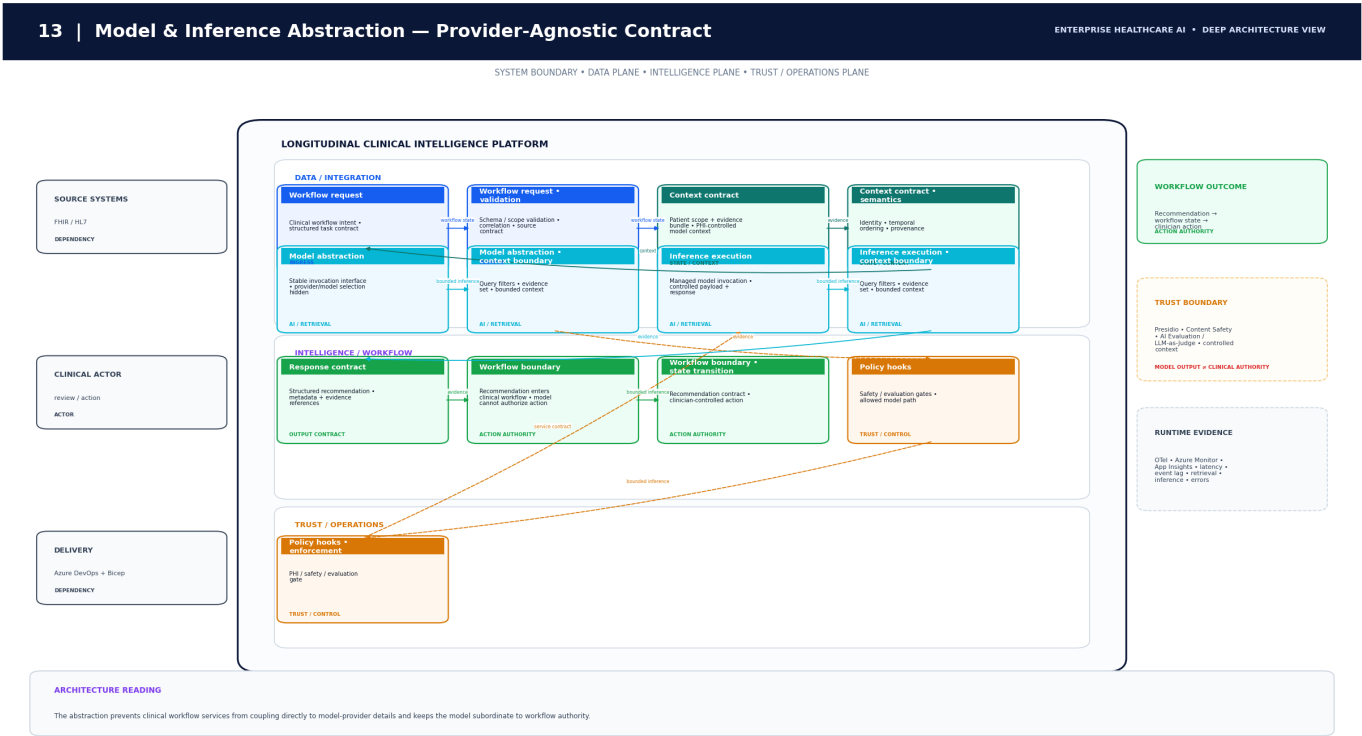
12 | Context Engineering & Token Economics

Deep architecture view



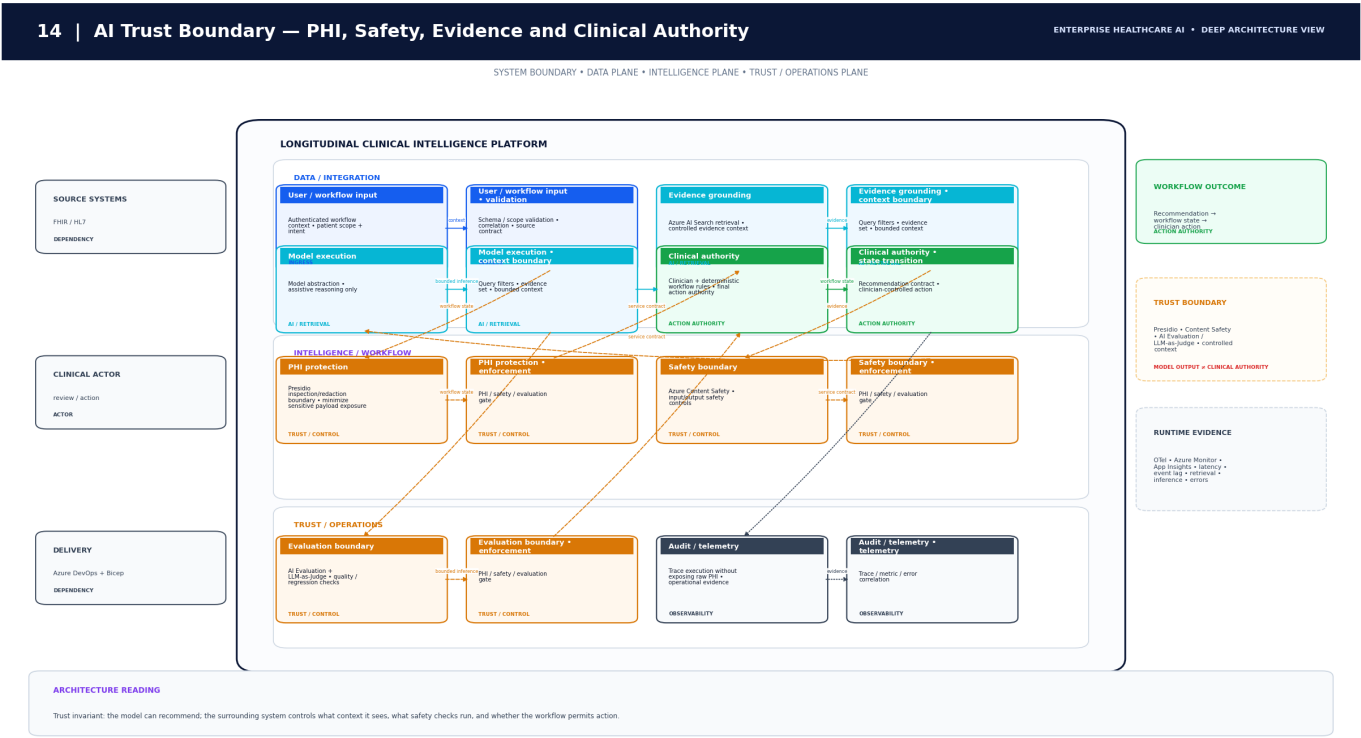
13 | Model & Inference Abstraction — Provider-Agnostic Contract

Deep architecture view



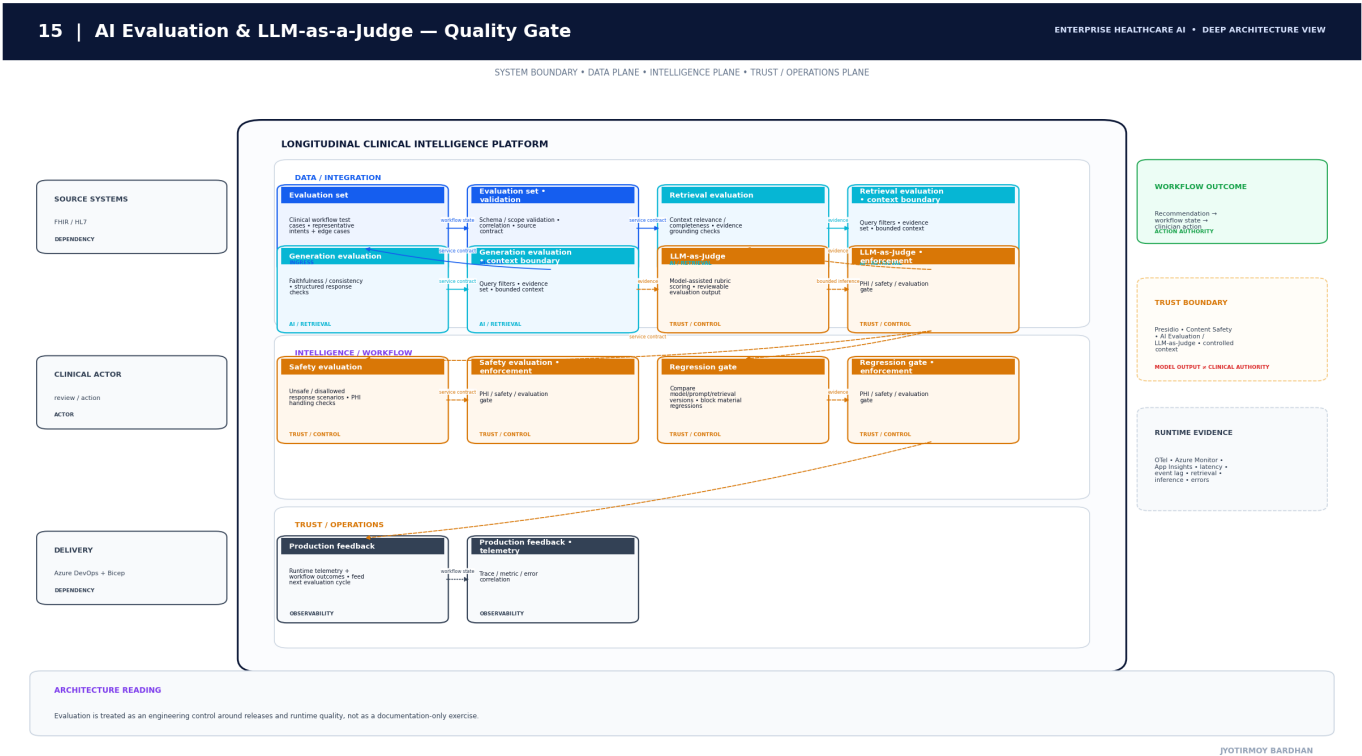
14 | AI Trust Boundary — PHI, Safety, Evidence and Clinical Authority

Deep architecture view



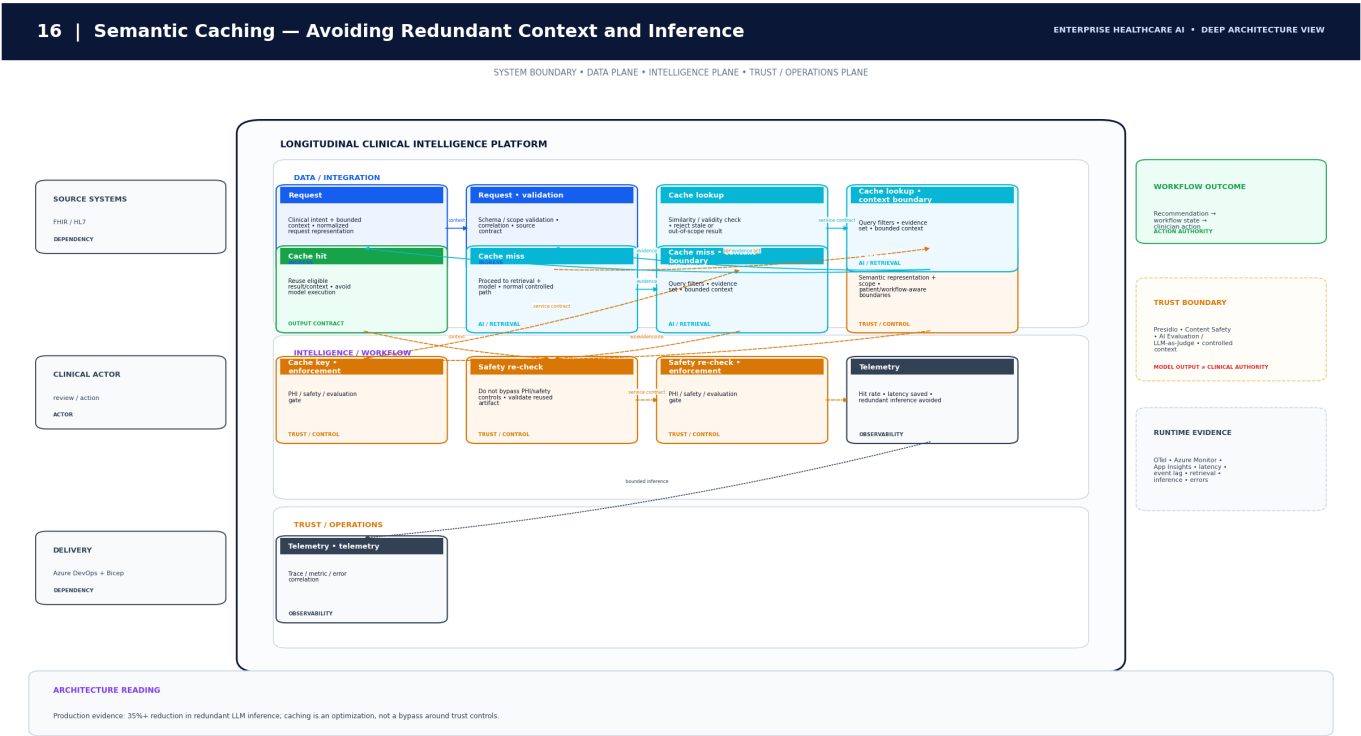
15 | AI Evaluation & LLM-as-a-Judge — Quality Gate

Deep architecture view



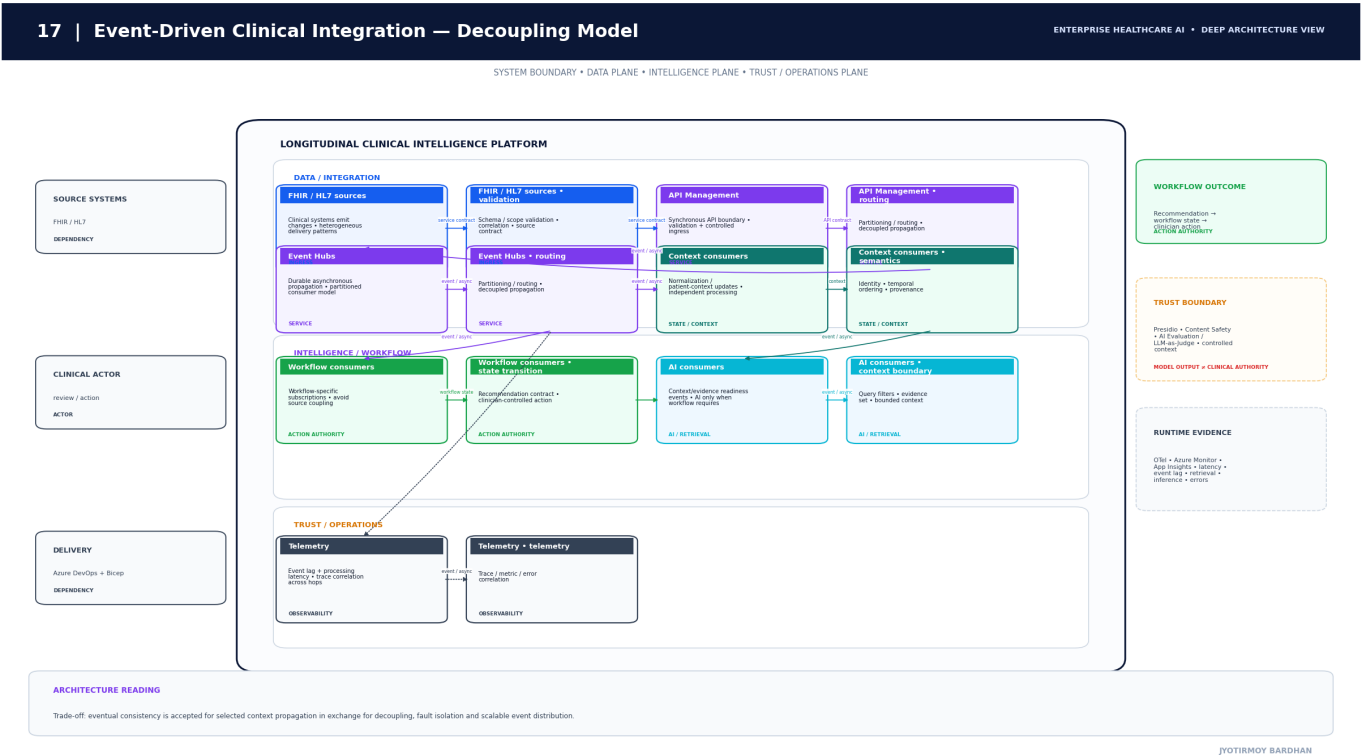
16 | Semantic Caching — Avoiding Redundant Context and Inference

Deep architecture view



17 | Event-Driven Clinical Integration — Decoupling Model

Deep architecture view

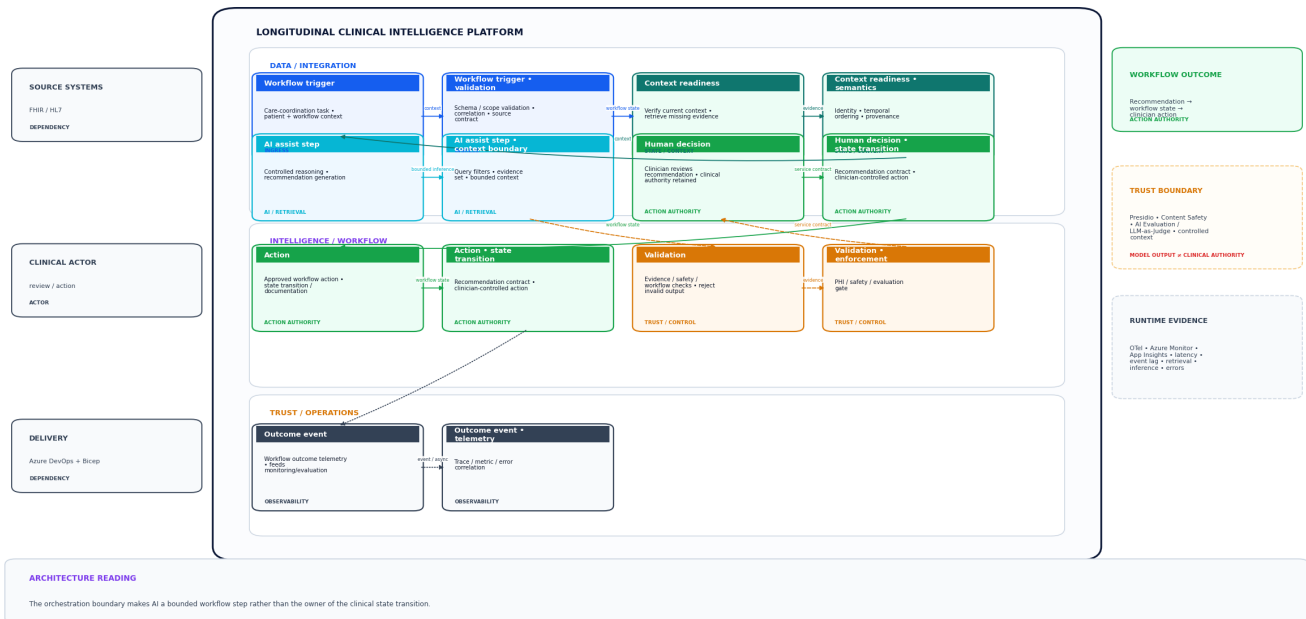


Deep architecture view

18 | Clinical Workflow Orchestration — AI as Assistive Step

ENTERPRISE HEALTHCARE AI • DEEP ARCHITECTURE VIEW

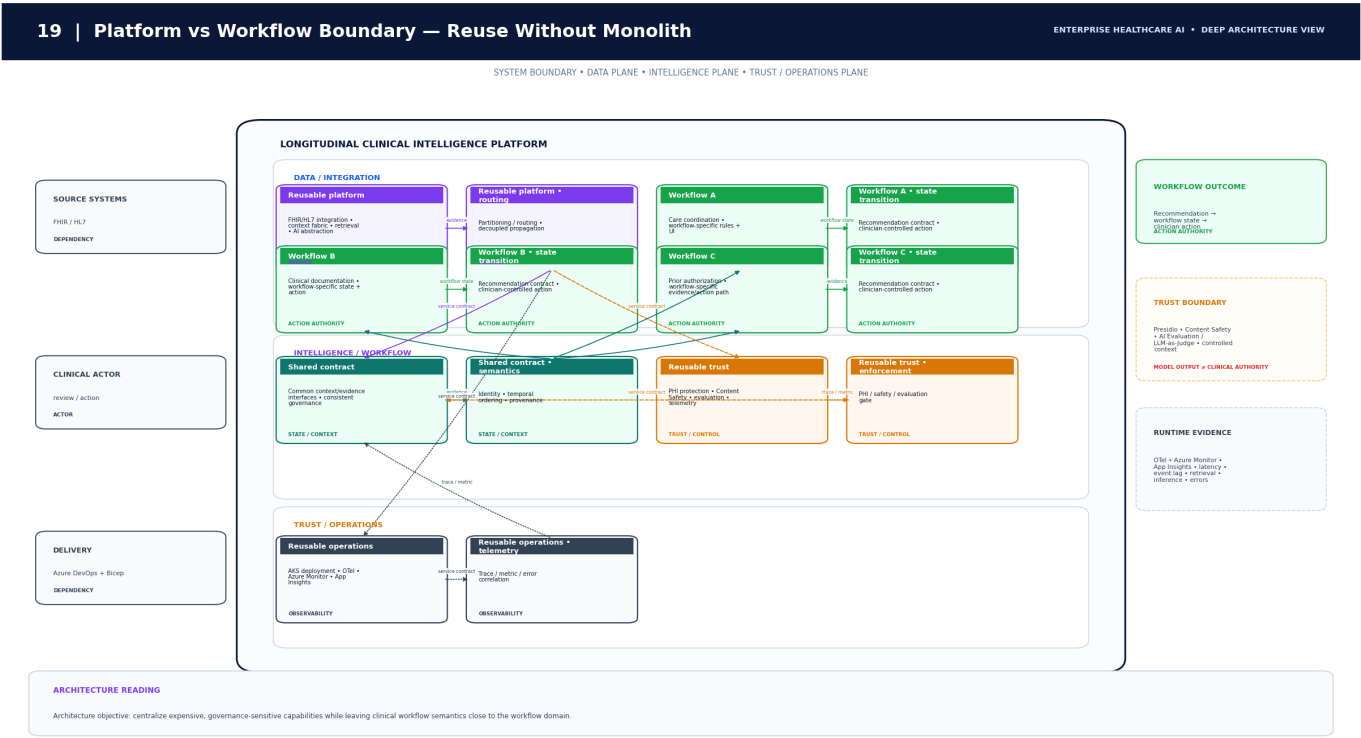
SYSTEM BOUNDARY • DATA PLANE • INTELLIGENCE PLANE • TRUST / OPERATIONS PLANE



Architecture reading: The orchestration boundary makes AI a bounded workflow step rather than the owner of the clinical state transition.

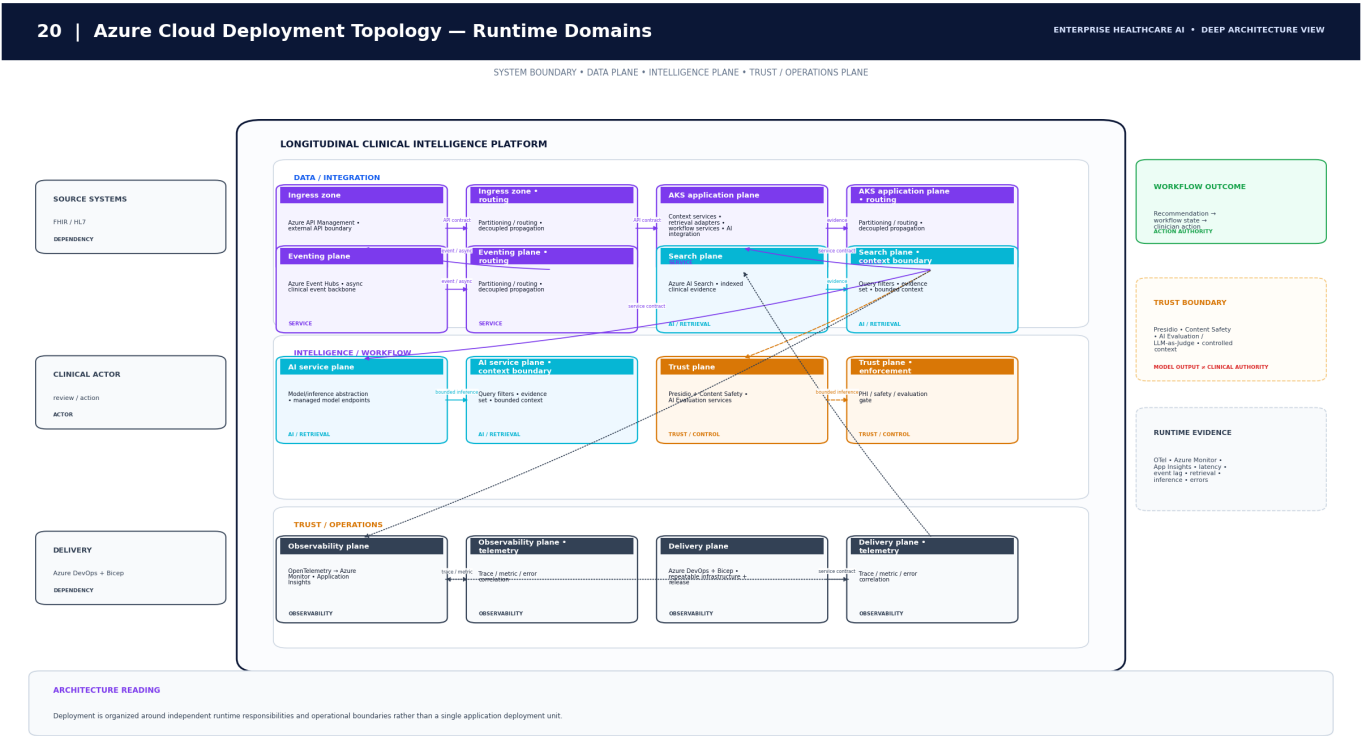
19 | Platform vs Workflow Boundary — Reuse Without Monolith

Deep architecture view



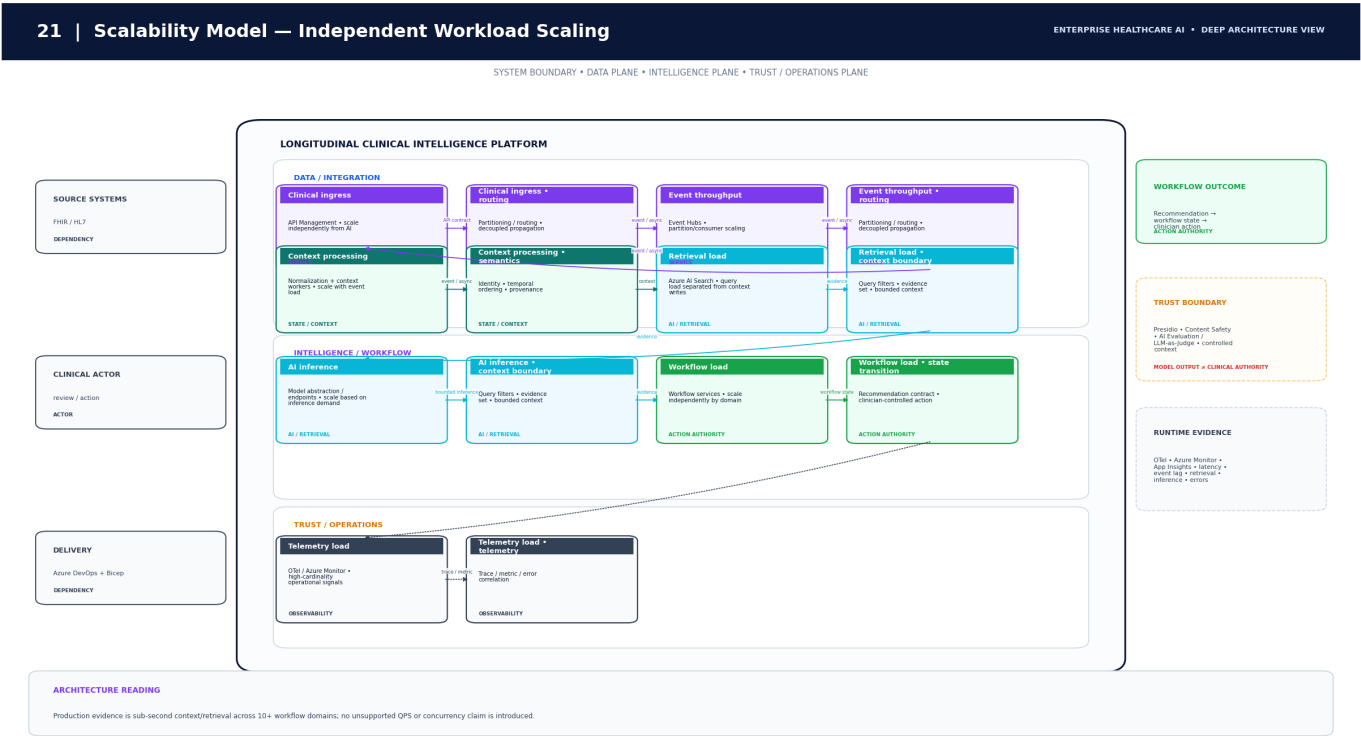
20 | Azure Cloud Deployment Topology — Runtime Domains

Deep architecture view



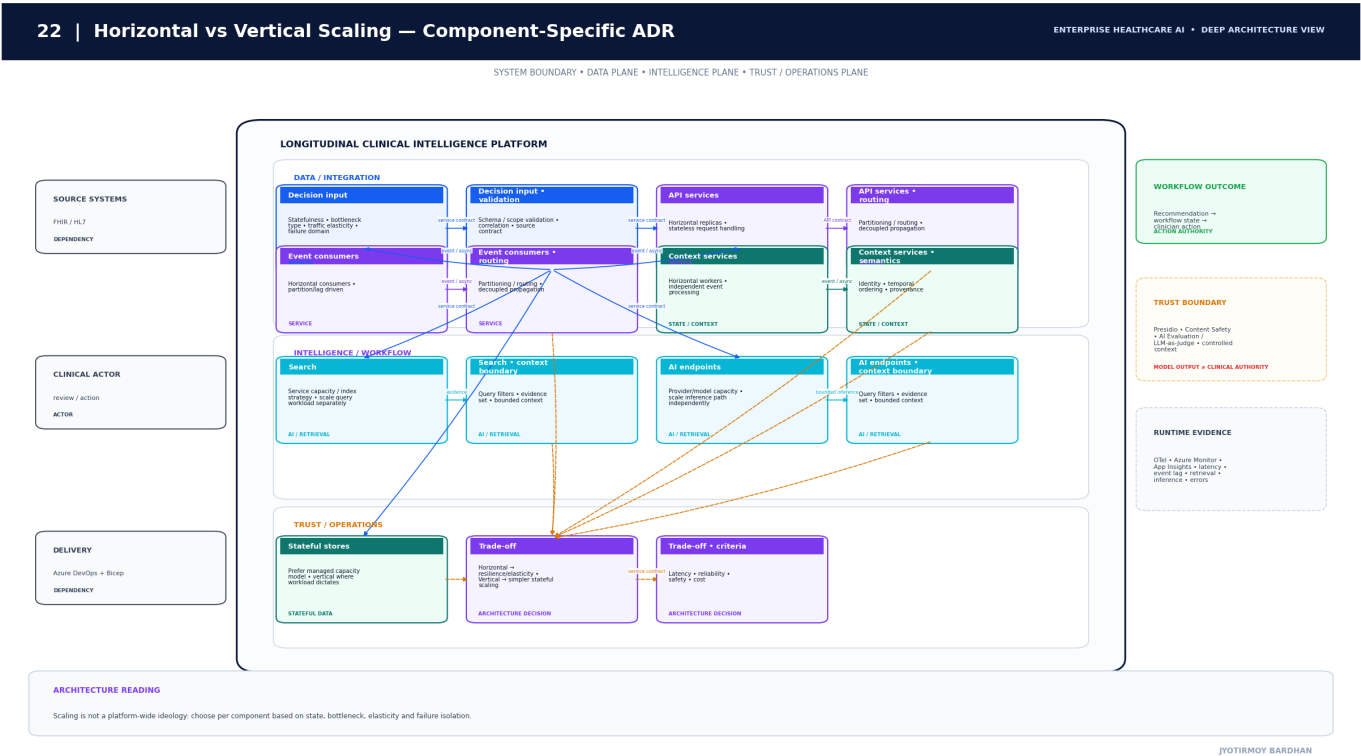
21 | Scalability Model — Independent Workload Scaling

Deep architecture view



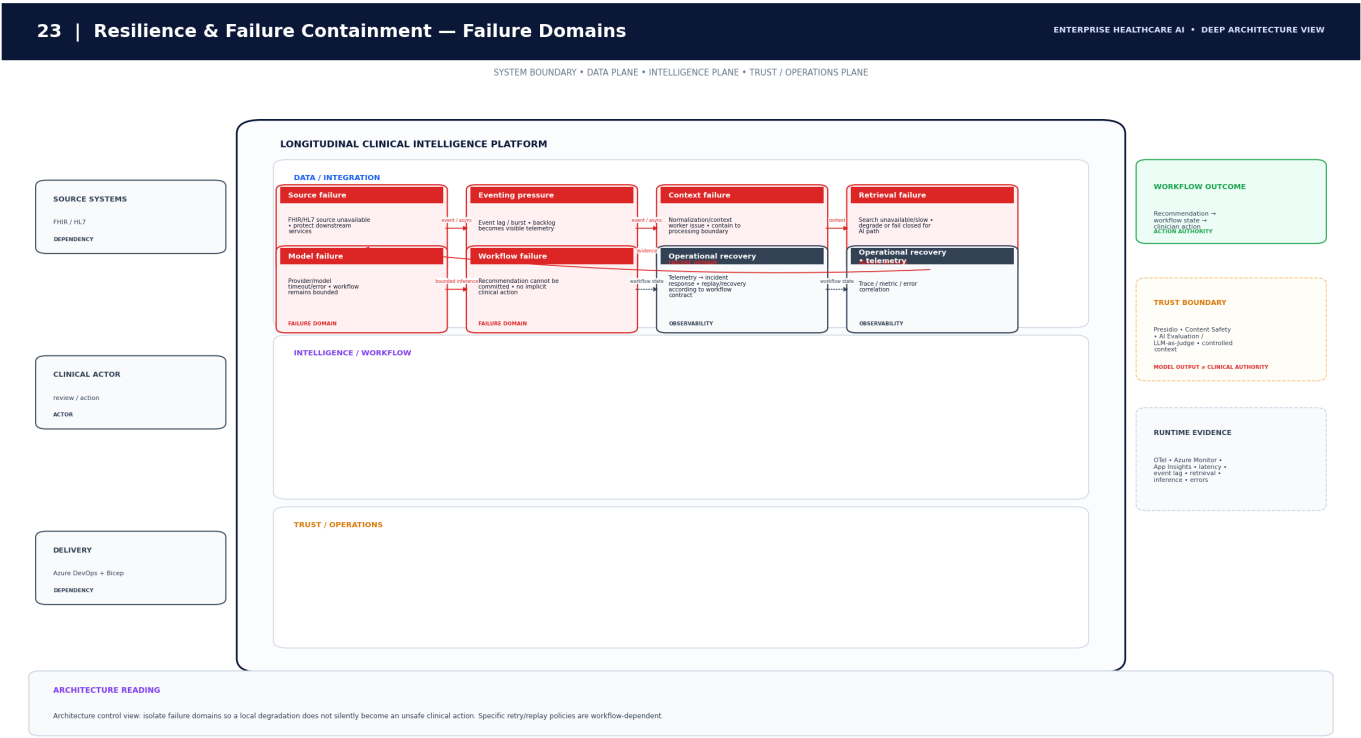
22 | Horizontal vs Vertical Scaling — Component-Specific ADR

Deep architecture view



23 | Resilience & Failure Containment — Failure Domains

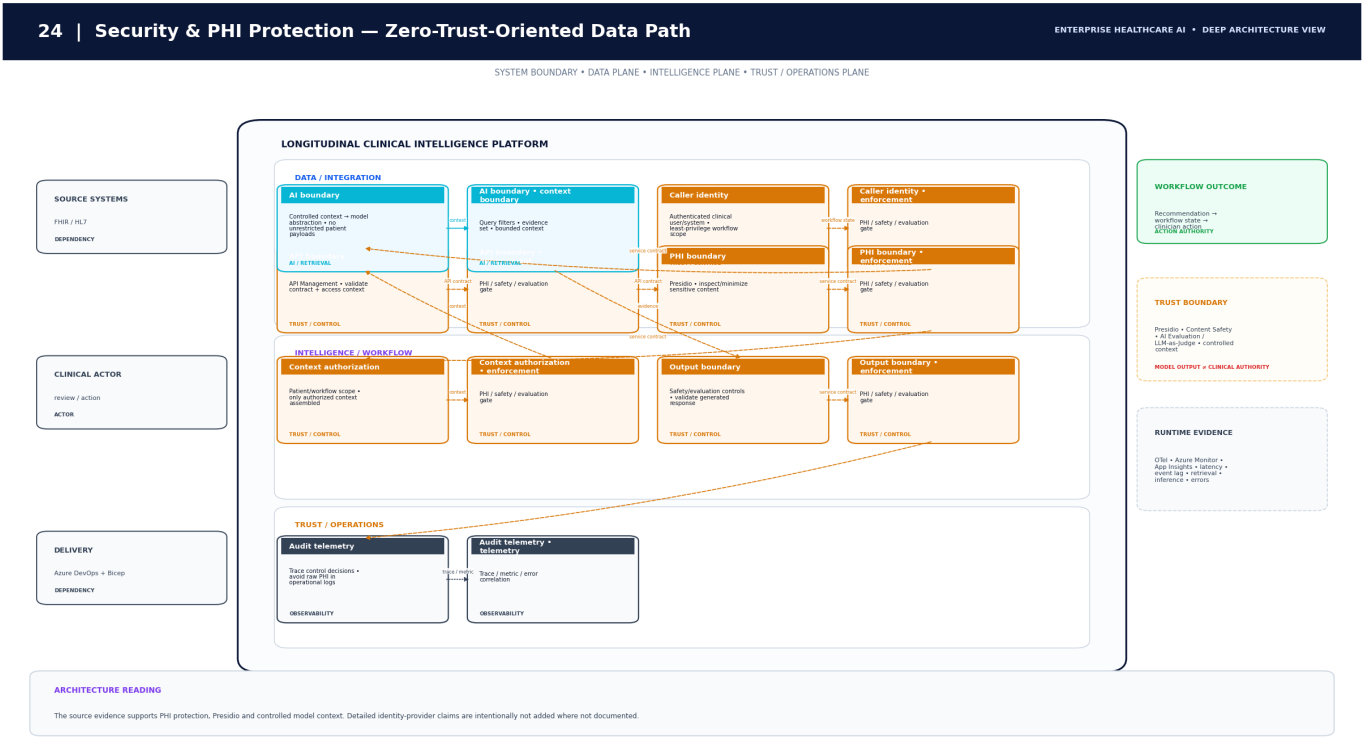
Deep architecture view



Architecture reading: Architecture control view: isolate failure domains so a local degradation does not silently become an unsafe clinical action. Specific retry/replay policies are workflow-dependent.

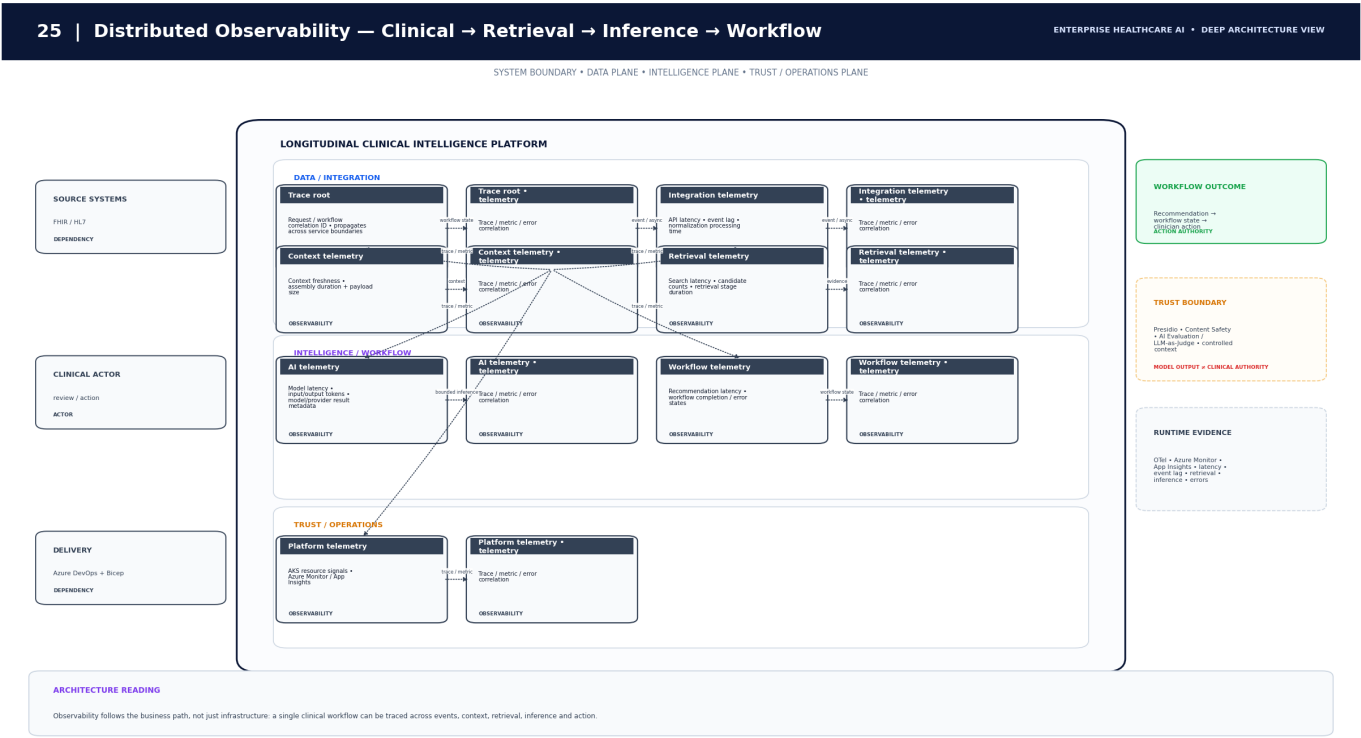
24 | Security & PHI Protection — Zero-Trust-Oriented Data Path

Deep architecture view



25 | Distributed Observability — Clinical → Retrieval → Inference → Workflow

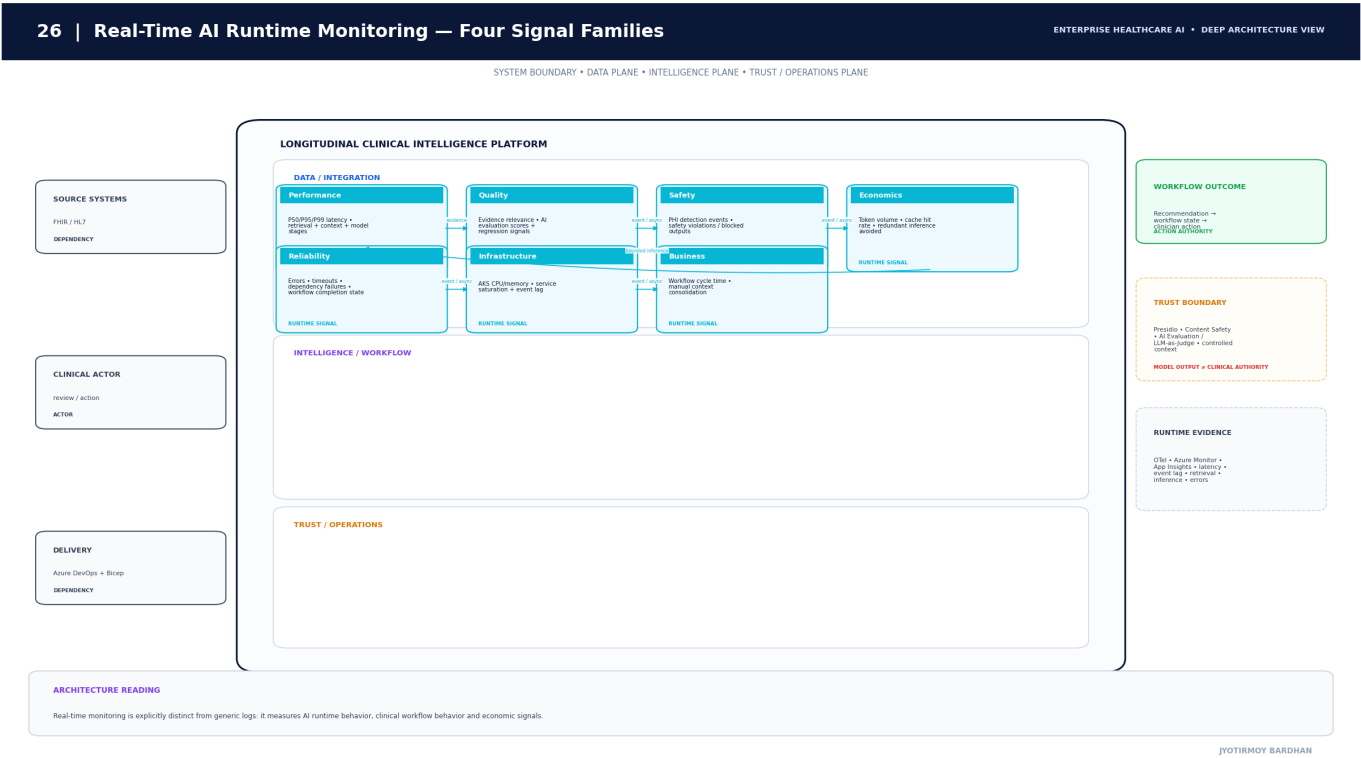
Deep architecture view



Architecture reading: Observability follows the business path, not just infrastructure: a single clinical workflow can be traced across events, context, retrieval, inference and action.

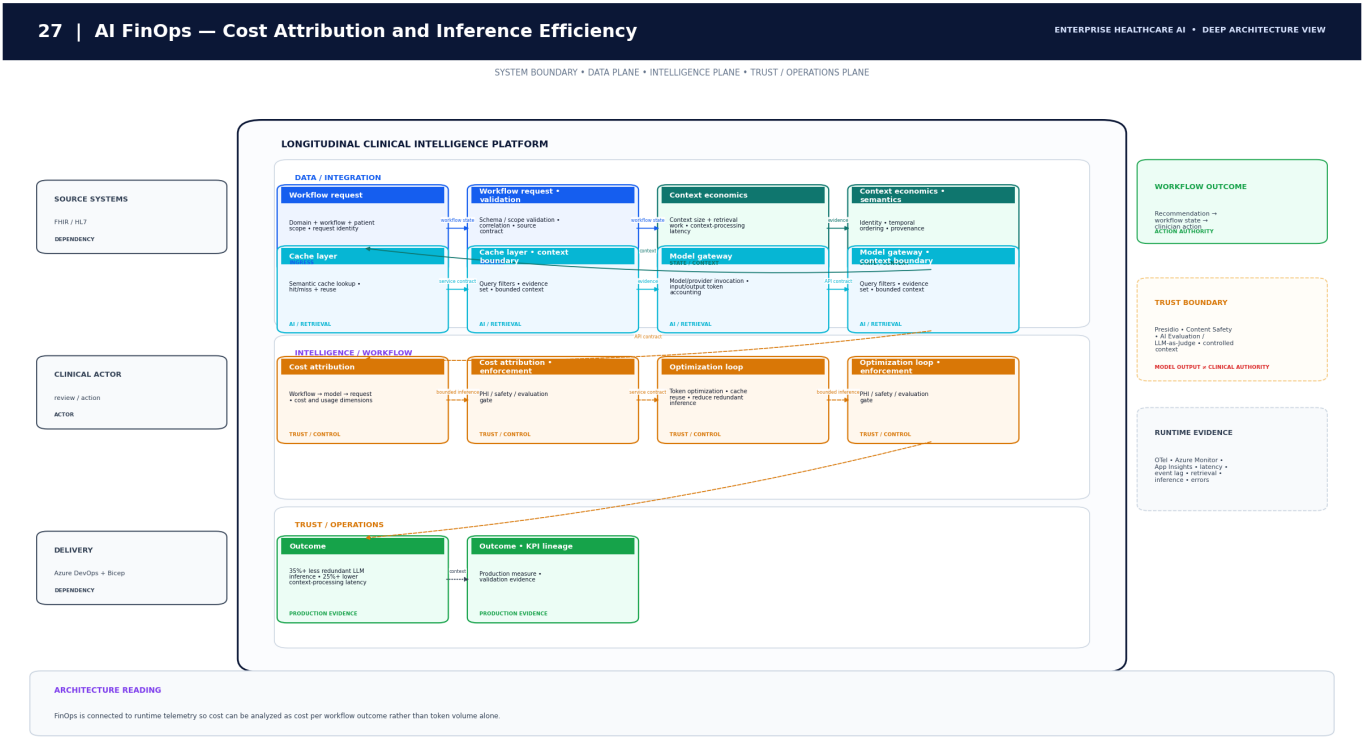
26 | Real-Time AI Runtime Monitoring — Four Signal Families

Deep architecture view



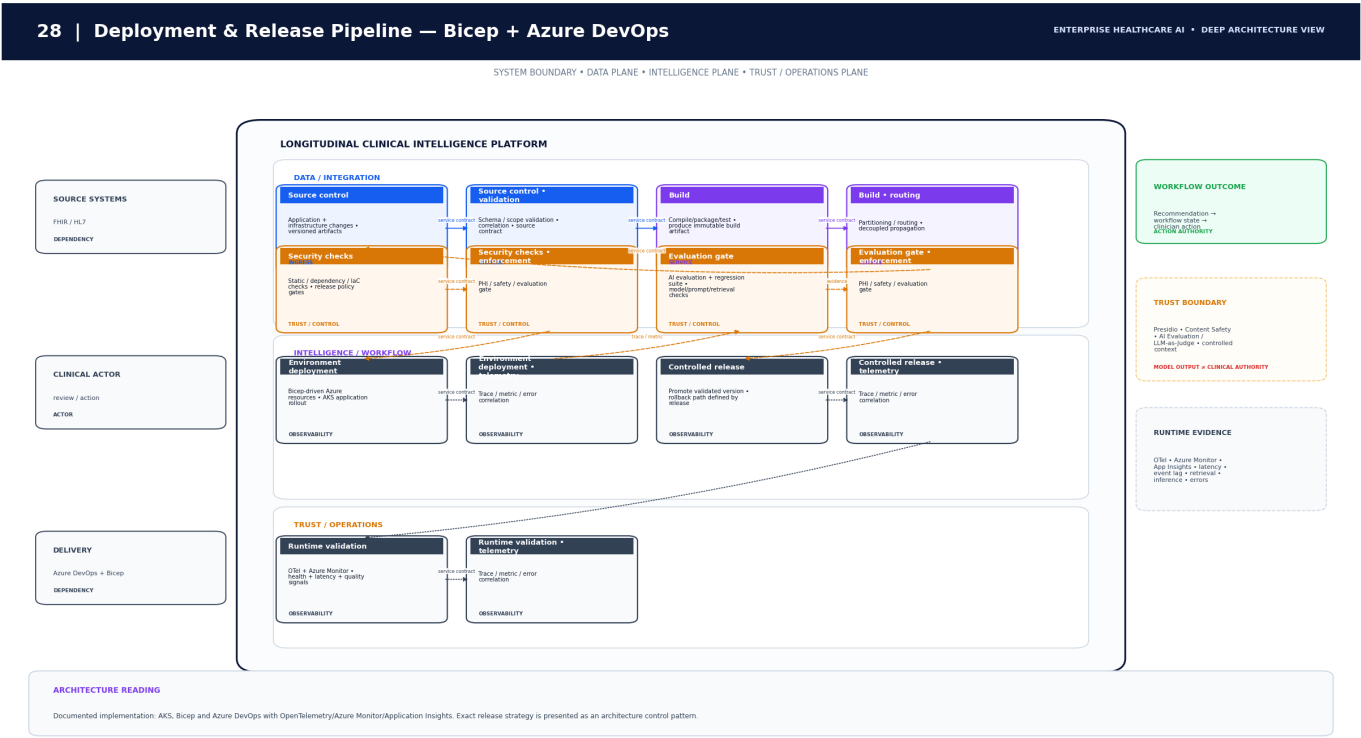
27 | AI FinOps — Cost Attribution and Inference Efficiency

Deep architecture view



28 | Deployment & Release Pipeline — Bicep + Azure DevOps

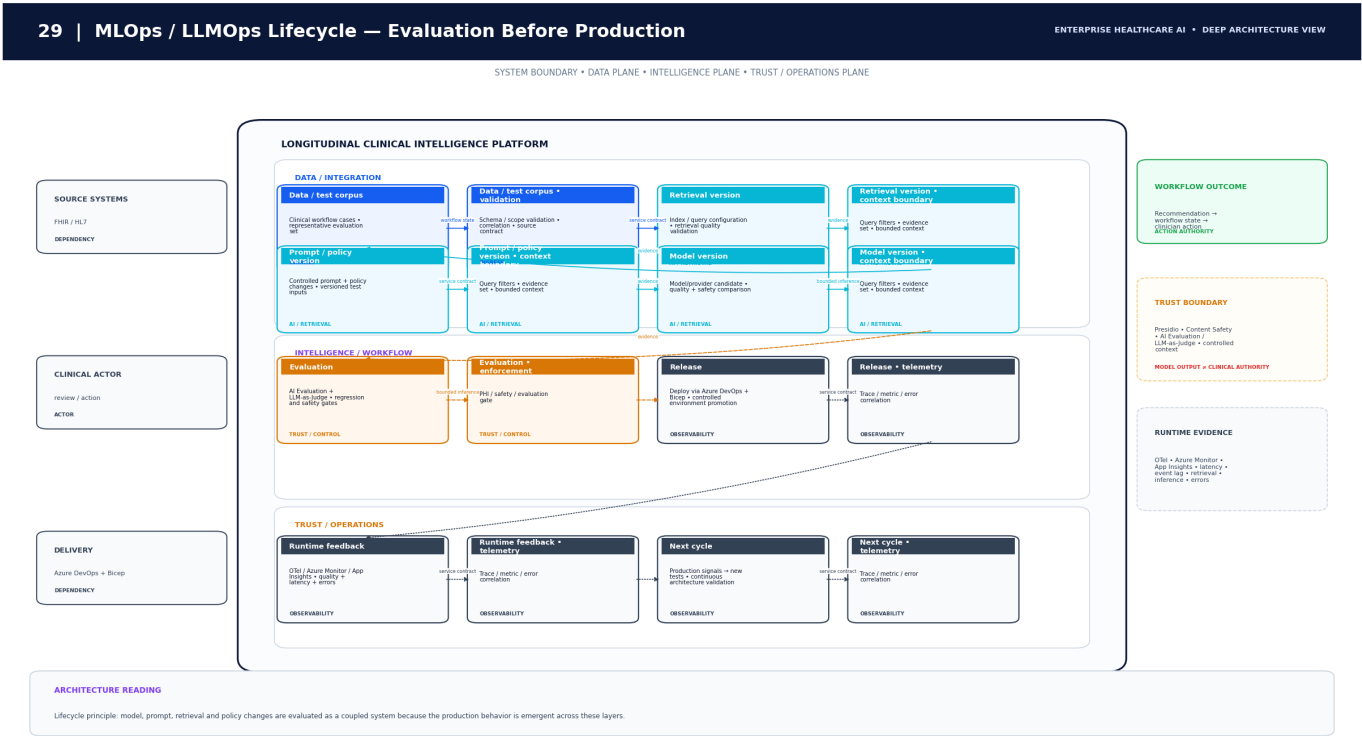
Deep architecture view



Architecture reading: Documented implementation: AKS, Bicep and Azure DevOps with OpenTelemetry/Azure Monitor/Application Insights. Exact release strategy is presented as an architecture control pattern.

29 | MLOps / LLMOps Lifecycle — Evaluation Before Production

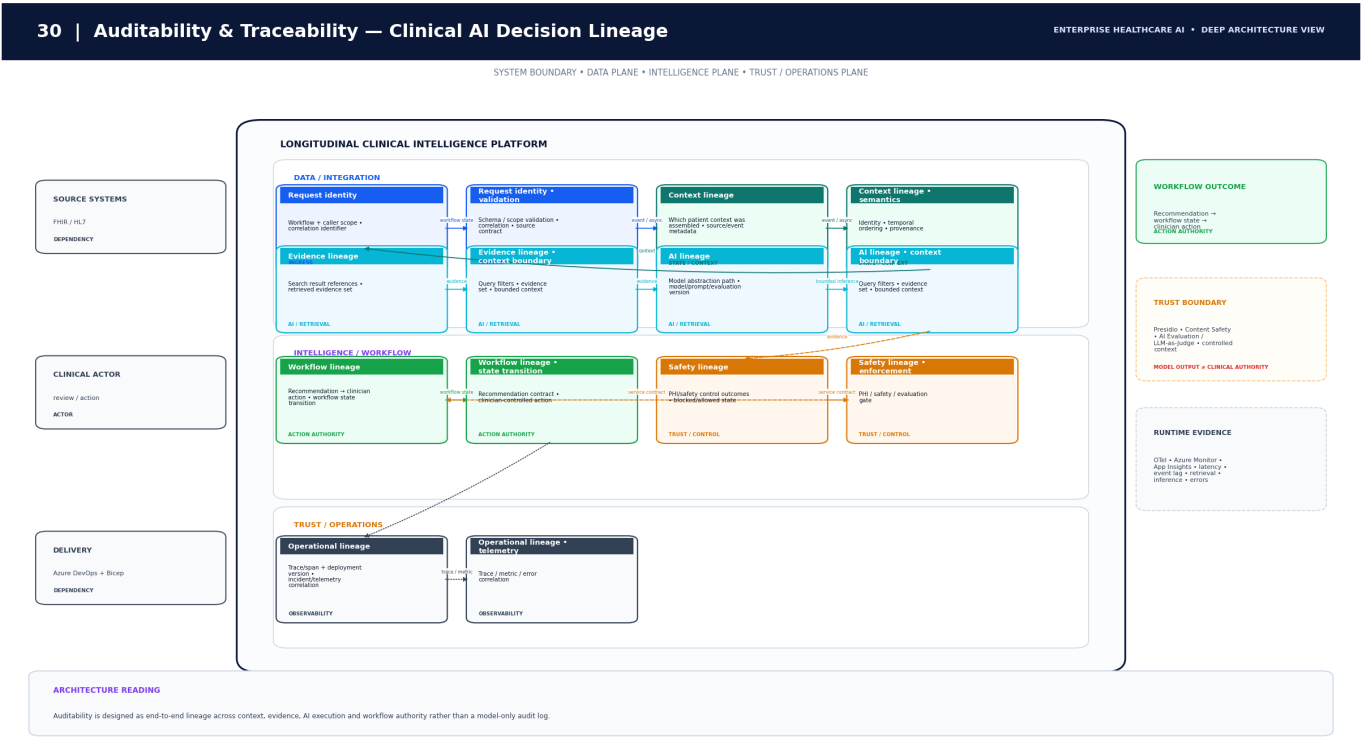
Deep architecture view



Architecture reading: Lifecycle principle: model, prompt, retrieval and policy changes are evaluated as a coupled system because the production behavior is emergent across these layers.

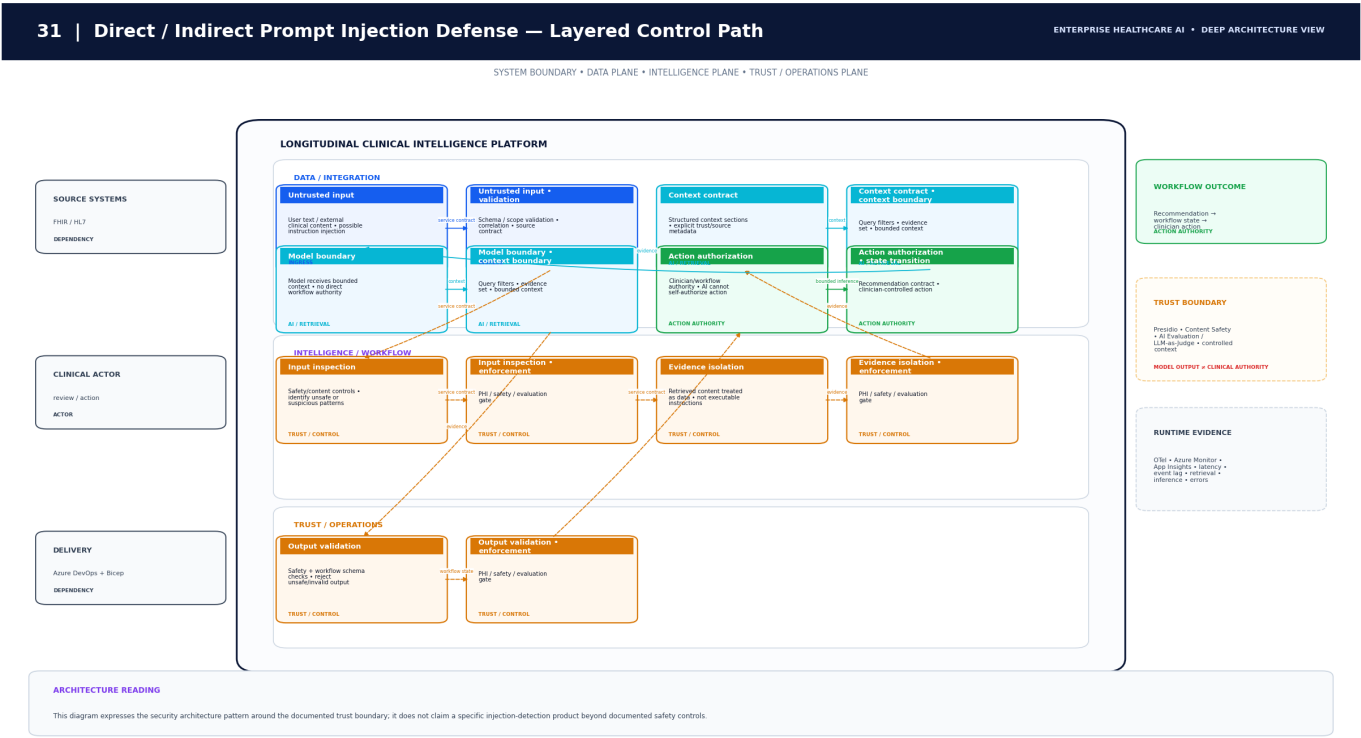
30 | Auditability & Traceability — Clinical AI Decision Lineage

Deep architecture view



31 | Direct / Indirect Prompt Injection Defense — Layered Control Path

Deep architecture view

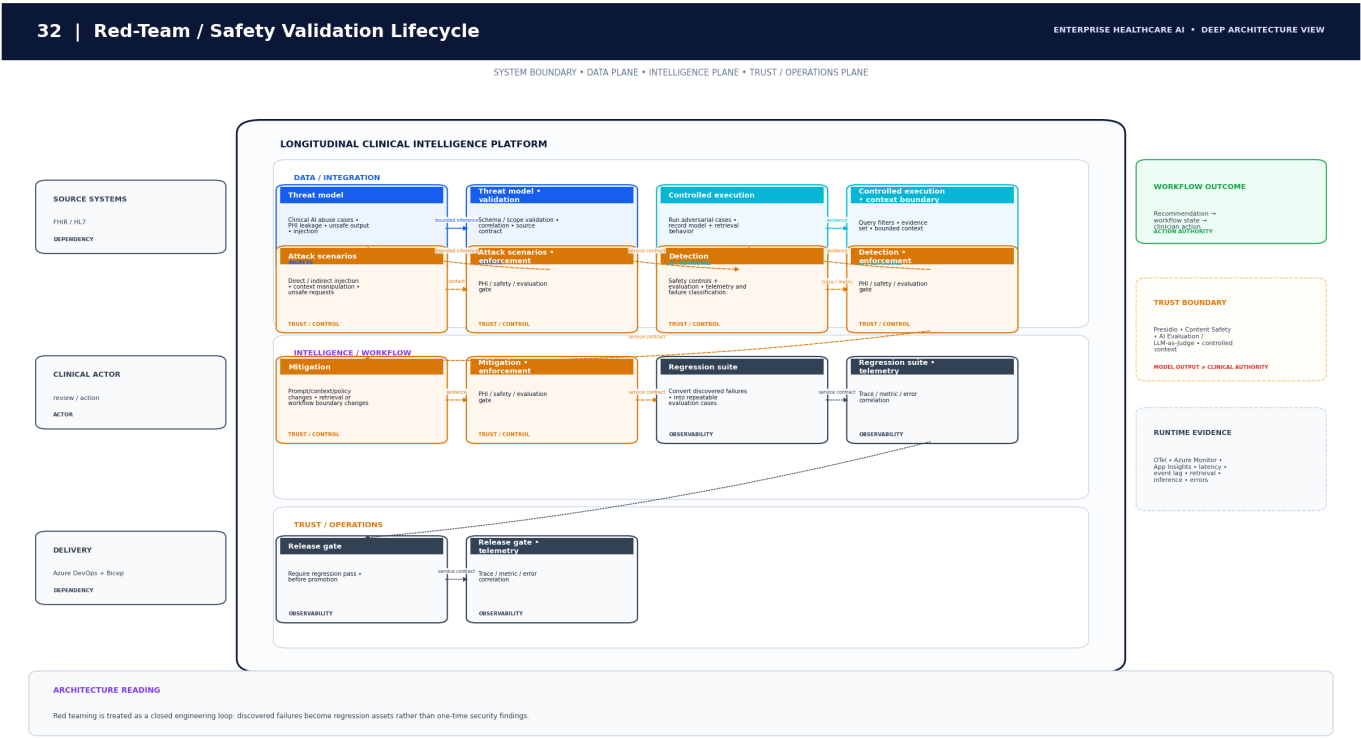


JYOTIRMOY BARDHAN

Architecture reading: This diagram expresses the security architecture pattern around the documented trust boundary; it does not claim a specific injection-detection product beyond documented safety controls.

32 | Red-Team / Safety Validation Lifecycle

Deep architecture view



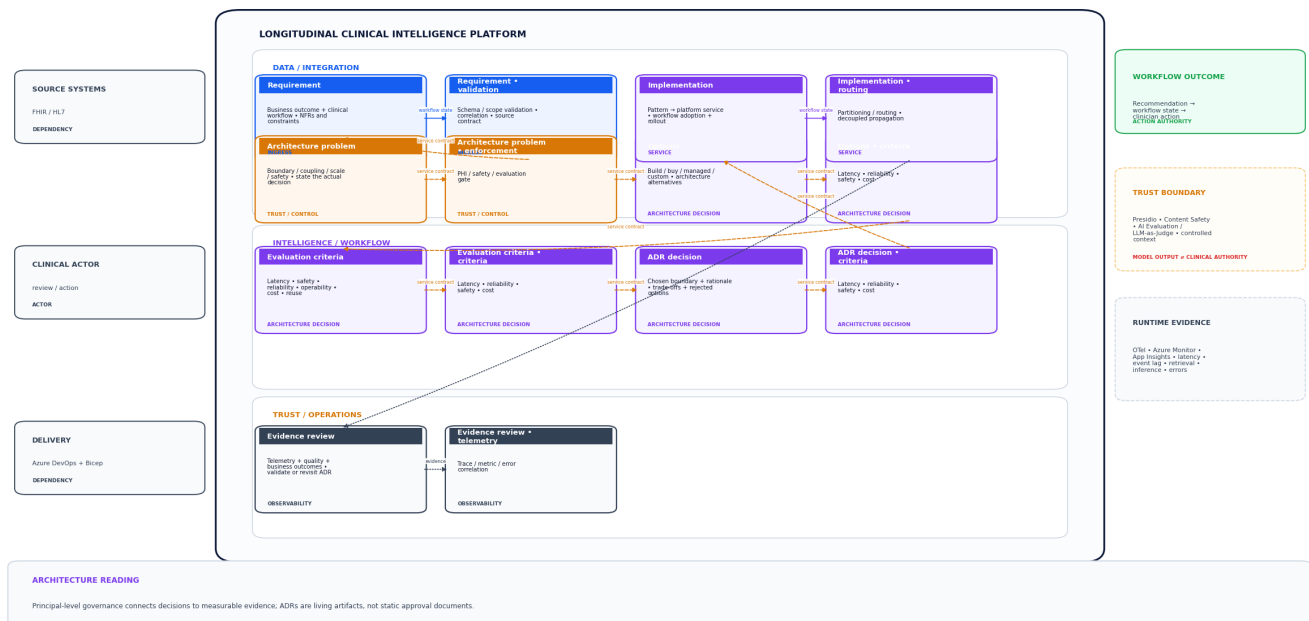
33 | ADR / Architecture Governance — Decision Operating Model

Deep architecture view

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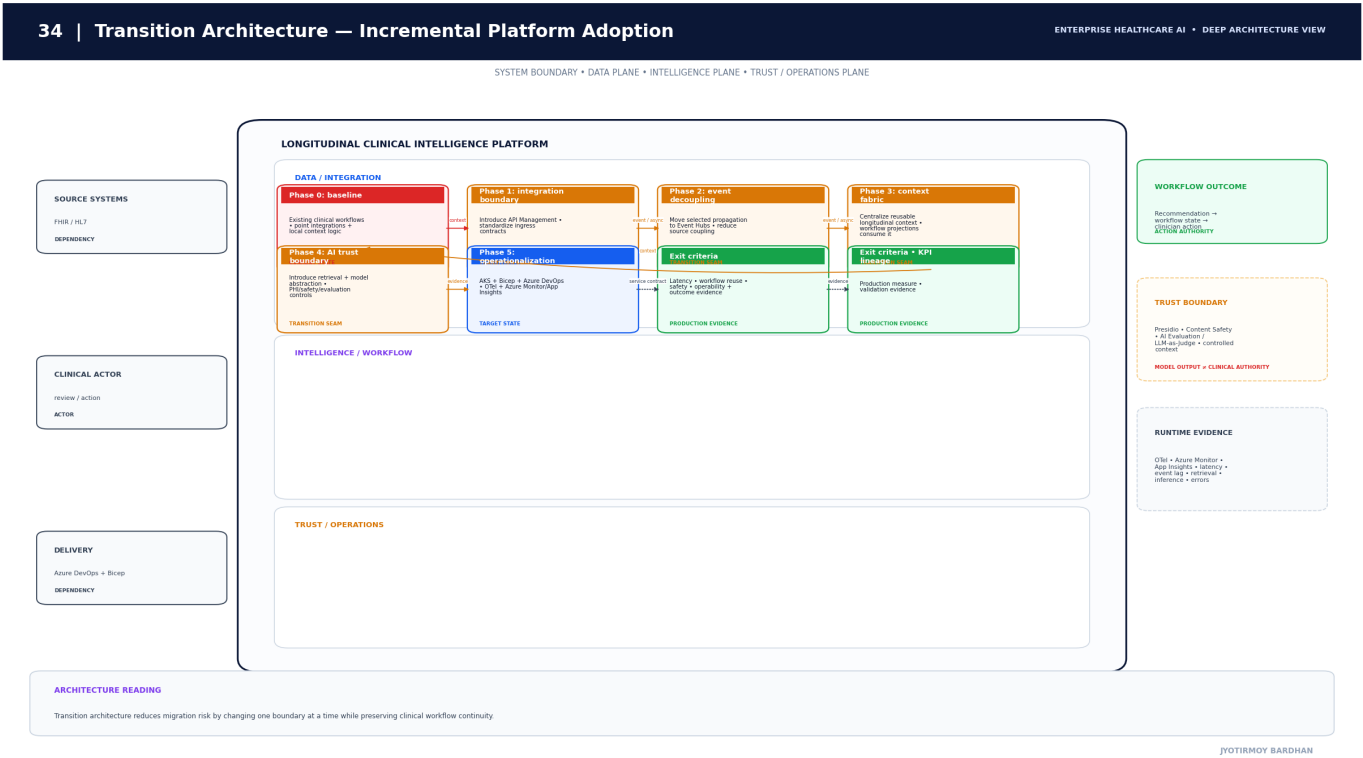
SYSTEM BOUNDARY • DATA PLANE • INTELLIGENCE PLANE • TRUST / OPERATIONS PLANE



Architecture reading: Principal-level governance connects decisions to measurable evidence; ADRs are living artifacts, not static approval documents.

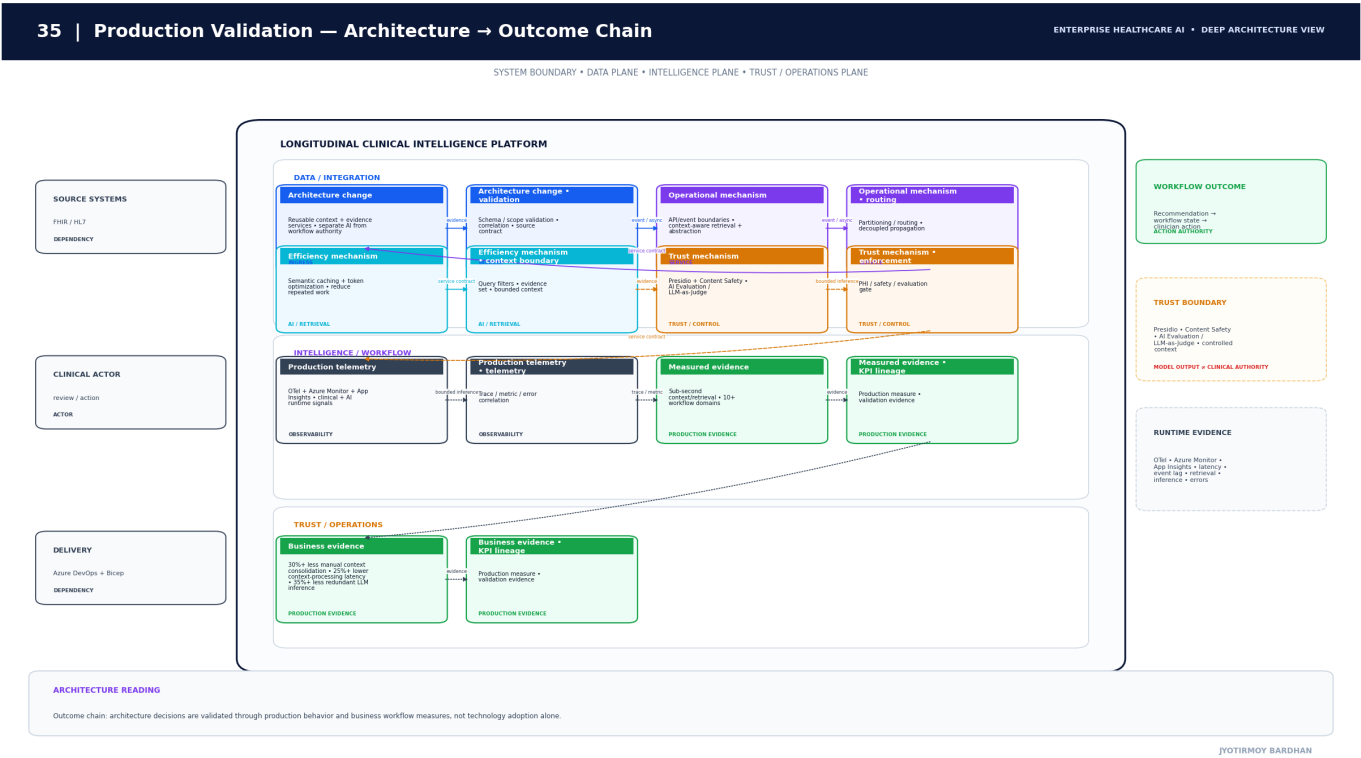
34 | Transition Architecture — Incremental Platform Adoption

Deep architecture view



35 | Production Validation — Architecture → Outcome Chain

Deep architecture view



Problem Framing

Fragmented clinical data, heterogeneous FHIR/HL7 semantics and repeated context consolidation created the need for reusable longitudinal context and governed AI integration.

Principal architecture lens

Architecture decisions are explicit boundaries with measurable operational consequences. The platform is validated through production behavior, operational evidence and workflow outcomes.

Real-World Constraints

Source variability, PHI protection, longitudinal state, retrieval latency, multiple workflow domains, safety and clinician control shape the architecture.

Principal architecture lens

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NFRs

Low retrieval/context latency, availability, fault isolation, secure PHI handling, traceability, reusable platform boundaries, controlled AI behavior and operational measurability.

Principal architecture lens

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Integration Boundary

Azure API Management is the documented ingress boundary; Azure Event Hubs provides asynchronous propagation where decoupling is beneficial.

Principal architecture lens

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Canonical Clinical Model

FHIR/HL7 representations are normalized before downstream context and retrieval services consume them; identity, timestamps and provenance are part of context semantics.

Principal architecture lens

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Patient Context Fabric

Longitudinal patient context is a reusable platform capability rather than duplicated inside each workflow.

Principal architecture lens

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Retrieval Architecture

Azure AI Search is the documented retrieval substrate. Retrieval is a measurable pipeline producing evidence for bounded context.

Principal architecture lens

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Context Engineering

Adaptive context assembly controls evidence entering model context; semantic caching and token optimization reduce repeated work.

Principal architecture lens

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Model / Inference Abstraction

Workflow services call a stable abstraction so provider-specific invocation details do not leak into clinical workflow contracts.

Principal architecture lens

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AI Trust Boundary

Presidio, Azure Content Safety and AI Evaluation / LLM-as-a-Judge sit around the AI execution boundary; workflow authority remains external to the model.

Principal architecture lens

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Semantic Caching

Caching is an efficiency mechanism; scope and safety controls remain applicable and cache reuse is not a trust bypass.

Principal architecture lens

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Event-Driven Runtime

Event Hubs decouples propagation from downstream context processing and absorbs source timing variability.

Principal architecture lens

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Workflow Authority

AI output is a recommendation/assistive artifact; workflow state transition and clinician action remain outside model execution.

Principal architecture lens

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Platform vs Workflow

Reusable platform capabilities include interoperability, context, retrieval, AI abstraction, safety/evaluation and observability; workflows retain domain-specific action semantics.

Principal architecture lens

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Azure Deployment

AKS hosts the documented runtime; Bicep and Azure DevOps provide the documented infrastructure/deployment path; managed services remain separate boundaries.

Principal architecture lens

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Scalability

Production evidence is sub-second context/retrieval across 10+ domains; the architecture separates ingress, event processing, context, retrieval, AI and workflow responsibilities.

Principal architecture lens

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Scaling Decision Logic

Stateless ingress/workers favor horizontal scaling; event consumers scale around throughput; managed services follow service capacity; stateful decisions remain component-specific.

Principal architecture lens

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Failure Containment

Source, event, context, retrieval, model and workflow concerns are separated into failure domains; a dependency failure must not silently become a clinical action.

Principal architecture lens

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PHI Protection

Presidio and controlled context minimize unnecessary patient-data exposure to model execution and operational telemetry.

Principal architecture lens

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Observability

OpenTelemetry, Azure Monitor and Application Insights provide distributed visibility across the clinical-to-AI path.

Principal architecture lens

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Real-Time AI Monitoring

Runtime monitoring correlates performance, quality, safety, reliability and utilization signals; documented evidence anchors include retrieval latency and inference reduction.

Principal architecture lens

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AI FinOps

Semantic caching and token optimization are documented efficiency mechanisms; 35%+ less redundant LLM inference is the measured evidence.

Principal architecture lens

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MLOps / LLMOps

Model, retrieval, context and evaluation behavior should be versioned and validated together; documented evaluation/deployment mechanisms close the lifecycle.

Principal architecture lens

Architecture decisions are explicit boundaries with measurable operational consequences. The platform is validated through production behavior, operational evidence and workflow outcomes.

CI/CD

Azure DevOps and Bicep provide the documented delivery path; AI behavior changes should be validated before promotion.

Principal architecture lens

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Auditability

Traceability connects request scope, patient context, retrieved evidence, AI execution metadata, trust-control outcomes and workflow state.

Principal architecture lens

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Prompt Injection / Red Teaming

Direct/indirect injection are trust-boundary risks; untrusted content is data, context is structured, output is validated, and workflow authority remains external.

Principal architecture lens

Architecture decisions are explicit boundaries with measurable operational consequences. The platform is validated through production behavior, operational evidence and workflow outcomes.

ADR Operating Model

ADRs record context, constraints, alternatives, evaluation criteria, decision, trade-offs and validation evidence.

Principal architecture lens

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Production Validation & Principal Contribution

The architecture links reusable context, evidence retrieval, inference abstraction, trust controls, efficiency mechanisms and distributed observability to measured workflow outcomes.

Principal architecture lens

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